Deep Jump Gaussian Processes for Surrogate Modeling of High-Dimensional Piecewise Continuous Functions

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Abstract

We introduce Deep Jump Gaussian Processes (DJGP), a novel method for surrogate modeling of a piecewise continuous function on a high-dimensional domain. DJGP addresses the limitations of conventional Jump Gaussian Processes (JGP) in high-dimensional input spaces by integrating region-specific, locally linear projections with JGP modeling. These projections employ region-dependent matrices to capture local subspace structures, making them well-suited to the inherently localized modeling behavior of JGPs, a variant of local Gaussian processes. To control model complexity, we place a Gaussian Process prior on the projection matrices, allowing them to evolve smoothly across the input space. The projected inputs are then modeled with a JGP to capture piecewise continuous relationships with the response. This yields a distinctive two-layer deep learning of GP/JGP. We develop a scalable variational inference algorithm to jointly learn the projection matrices and JGP hyperparameters. Rigorous theoretical analysis and extensive empirical studies are provided to justify the proposed approach. In particular, we derive an oracle error bound for DJGP. Experiments on synthetic and benchmark datasets demonstrate that DJGP achieves superior predictive accuracy and more reliable uncertainty quantification compared with existing methods.

Keywords: Gaussian process, piecewise continuous surrogates, locally linear projection, variational inference

1 Introduction

This paper addresses surrogate modeling of piecewise continuous system responses in high-dimensional input spaces. In many engineering and scientific domains, system responses can exhibit abrupt jumps or sharp transitions under small input perturbations. For instance, in geostatistics, subsurface rock properties such as porosity and permeability can change dramatically at sedimentary interfaces, naturally giving rise to piecewise continuous behavior (Chiles & Delfiner 2012). In materials science, first-order phase transitions (e.g., the ferromagnetic–paramagnetic shift at the Curie point) induce discontinuous changes in properties like magnetization and density (Park et al. 2022). In econometrics, regression discontinuity designs leverage sharp outcome changes at policy thresholds or eligibility cutoffs to identify causal effects (Kang et al. 2019). In smart manufacturing systems, system performance may change abruptly as operating conditions approach capacity constraints (Park et al. 2025). Developing surrogate models for piecewise continuous response surfaces is essential for data-driven understanding and uncertainty quantification of such systems.

While Gaussian processes (GPs) offer a flexible Bayesian nonparametric surrogate model with uncertainty quantification, they typically rely on stationary kernels such as the squared exponential, which induce strong correlations between nearby inputs and impose smoothness, making them poorly suited for modeling abrupt changes or discontinuities (Park 2022). Nonstationary GP models can better adapt these changes by adjusting their hyperparameters locally to capture varying covariance structures (Sampson & Guttorp 1992, Sauer et al. 2023*a*). Representative approaches include heteroskedastic GPs (Kersting et al. 2007, Quadrianto et al. 2009) and latent GPs that model kernel parameters such as variance or lengthscale (Paciorek & Schervish 2003, Tolvanen et al. 2014, Heinonen et al. 2016). A more flexible alternative is Deep Gaussian Processes (DGPs) (Lawrence & Moore 2007, Damianou & Lawrence 2013), which stack multiple GP layers to warp inputs into nonlinear feature spaces and map them to responses. This hierarchical structure enables DGPs to capture complex nonstationary patterns that shallow GPs cannot represent, but that comes with the cost of intractable inferences. Considerable effort has gone into scalable inference for DGPs, including Vecchia approximations (Sauer et al. 2023*b*), variational frameworks (Titsias 2009, Hensman et al. 2013, Damianou et al. 2011, Damianou 2015), and sampling-based methods (Havasi et al. 2018). Hybrid models that combine neural network layers with GP layers further improve flexibility and scalability (Dai et al. 2015, Wilson et al. 2016*b*,*a*, Lee et al. 2017). Despite these advances, most nonstationary GP models—including DGP-based variants—remain fundamentally smooth and tend to blur discontinuities. Additionally, DGPs are often data-hungry and require large training sets.

An approach directly suited for modeling piecewise continuous surrogates is the partitioned GP, which divides the input space into regions and fits an independent GP within each region. When the partitions align well with discontinuities, partitioned GPs can effectively represent piecewise continuous surrogates. To control model complexity and computational cost, existing methods typically constrain how the space is partitioned. Common approaches include tessellation-based methods (e.g., Voronoi diagrams) (Kim et al. 2005, Pope et al. 2021, Luo et al. 2021) and treed partitioning (Gramacy & Lee 2008, Konomi et al. 2014, Taddy et al. 2011). These approaches improve scalability but rely on simplistic domain partitioning, making them less effective for complex or nonlinear boundaries.

A recent and more flexible approach is the Jump Gaussian Process (JGP) (Park 2022). Rather than explicitly modeling a global partition of the input space, JGP constructs local approximations of the partition boundaries. If the boundary is sufficiently smooth, it can be locally approximated by a linear or low-order polynomial function. At each test location, JGP fits both the local polynomial boundary and the local GP parameters using data with a small neighborhood of the test location. By leveraging these local approximations, JGP can represent piecewise continuous surrogates with highly complex regional boundaries. However, JGP faces challenges in high-dimensional settings. As input dimensionality increases, data sparsity grows, requiring larger neighborhoods to obtain sufficient local training data. This leads to coarser approximations and ultimately limits JGP's ability to capture fine-grained local structures in high-dimensional spaces.

The limitations of JGP motivate us to investigate dimensionality reduction for JGP. An easy fix of the dimensionality issue may be to apply dimensionality reduction prior to GP modeling, e.g. linear technique such as Principal Component Analysis (PCA)(Abdi & Williams 2010), nonlinear technique such as kernel PCA(Schölkopf et al. 1997), Isomap (Balasubramanian & Schwartz 2002), local linear embedding (Roweis & Saul 2000), autoencoders (Wang et al. 2016), or t-SNE (Maaten & Hinton 2008). However, these methods are unsupervised, meaning that they rely only on input data and ignore correlations between transformed features and the response variable. A better approach can be to use supervised approaches such as Sliced Inverse Regression (SIR) (Li 1991), supervised autoencoders(Makhzani & Frey 2015, Le et al. 2018) and conditional variational autoencoders (CVAEs) (Sohn et al. 2015, Kingma et al. 2014). Nevertheless, these supervised dimension-reduction techniques are still not optimized from the downstream GP modeling task.

Dimensionality reduction can be optimized directly for a target GP modeling task. For instance, the Mahalanobis Gaussian Process (AUEB & Lázaro-Gredilla 2013, MGP) learns a linear projection of the inputs to a low dimensional space, where the linear projection matrix is optimized jointly with the GP model parameters. The Gaussian Process Latent

Variable Model (Titsias & Lawrence 2010, GP-LVM) generalizes the linear projection with a nonlinear projection represented by a GP model, resulting in two-layer GP model with the first layer for non-linear feature mapping and the second layer for mapping to the response variable. The Deep Mahalanobis Gaussian Process (de Souza et al. 2022, DMGP) extends MGP with a similar two-layer design. It introduces a distinct linear projection matrix for each input location, with GP priors enforcing smooth variation of these matrices across the input space. The projected features are passed to another GP layer to model the response.

Nevertheless, existing built-in dimensionality reduction methods are not tailored for JGP. Their integration is nontrivial due to a fundamental modeling difference: conventional GP models follow an inductive learning paradigm, whereas JGP operates in a transductive setting—fitting a local model at each test location. The goal of this paper is to develop a built-in dimensionality reduction approach specifically designed for JGP, yielding a piecewise continuous surrogate model that remains effective in high-dimensional input spaces.

Our approach employs locally linear projections to map high-dimensional inputs to low-dimensional latent features. For each test location, we define a location-specific linear projection matrix, with spatial correlations across these matrices enforced via a GP prior. The resulting local latent features are then mapped to the response using a JGP model, yielding a unique two-layer GP/JGP architecture, which is referred to as Deep Jump Gaussian Processes (DJGP).

The remainder of this paper is organized as follows. Section 2 reviews background on stationary GPs, jump GPs, and Mahalanobis GPs. Section 3 introduces the proposed DJGP model and its variational inference scheme, while Section 3.2 details the full inference procedure. Section 4 presents theoretical results, including prediction error and risk bounds. Section 5 provides numerical evaluations on synthetic datasets, along with a comprehensive hyperparameter sensitivity analysis. Section 6 reports results on real-world datasets. Finally, Section 7 concludes this paper.

2 Review

Here we provide a brief technical review to the key technical components: Stationary GP, Jump GP and Mahalanobis GP.

2.1 Stationary Gaussian Process (GP) Surrogates

Consider an unknown function $f : \mathcal{X} \to \mathbb{R}$ to relate an input $\boldsymbol{x} \in \mathcal{X}$ to a real response y, where $\mathcal{X} \subset \mathbb{R}^D$ denote the input space. We can build a stationary GP surrogate to f given its noisy evaluations, $y_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}\big(f(\boldsymbol{x}_i),\, \sigma^2\big),\ i = 1, \ldots, N$, where a prior distribution over f is defined by the stationary Gaussian process with a constant mean function μ and covariance kernel $c(\cdot,\cdot),\ f(x) \sim \mathcal{GP}\big(\mu, c(\cdot,\cdot;\theta)\big)$. The covariance kernel is a positive definite function $c : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ parameterized by hyperparameters θ. A common modeling assumption on the kernel is stationarity, where the covariance kernel depends only on the relative distance between inputs. A widely used stationary kernel is the squared exponential (SE):

$$k_{\text{SE}}(\boldsymbol{x}_i, \boldsymbol{x}_j; \sigma_f, \ell_1, \ldots, \ell_D) = \sigma_f^2 \exp\left(-\tfrac{1}{2}\sum_{m=1}^{D} \frac{(x_{im} - x_{jm})^2}{\ell_m^2}\right), \tag{1}$$

where x_{im} denotes the mth dimension of $\boldsymbol{x}_i$. The Matérn class provides greater flexibility by controlling the smoothness of the function (Wendland 2004).

Under this prior, the vector of the observed outputs $\boldsymbol{y}_N = [y_1, \ldots, y_N]^\top$ follows a multivariate normal distribution, $\boldsymbol{y}_N \sim \mathcal{N}\big(\mu\mathbf{1}_N,\, \sigma^2\boldsymbol{I}_N + \boldsymbol{C}_N\big)$, where $\boldsymbol{C}_N$ is a $N \times N$ matrix with its (i,j)th element equal to $c(\boldsymbol{x}_i, \boldsymbol{x}_j; \theta)$, $\mathbf{1}_N$ is a N-dimensional column vector of ones, and $\boldsymbol{I}_N$ is a N-dimensional identity matrix.

The hyperparameters θ, mean parameter μ, and noise variance σ^2—can be learned by maximizing the log marginal likelihood:

$$L(\theta, \mu, \sigma^2) = -\tfrac{1}{2}(\boldsymbol{y}_N - \mu\mathbf{1}_N)^\top\big[\sigma^2\boldsymbol{I}_N + \boldsymbol{C}_N\big]^{-1}(\boldsymbol{y}_N - \mu\mathbf{1}_N) - \tfrac{1}{2}\log\big|\sigma^2\boldsymbol{I}_N + \boldsymbol{C}_N\big| - \tfrac{N}{2}\log(2\pi),$$

using either gradient-based optimization or EM-style iterative schemes, depending on the model setting (Santner et al. 2003, Gramacy 2020, Titsias 2009). We use the hat notations $\hat{\mu}, \hat{\theta}, \hat{\sigma}^2$ to denote the estimated parameters.

Given the parameter estimates, we can derive the predictive distribution of f at a test input $\boldsymbol{x}_* \in \mathcal{X}$. The joint distribution of $\boldsymbol{y}_N$ and an unknown testing output $y(\boldsymbol{x}_*)$ is a multivariate normal (MVN) distribution. Applying the simple Gaussian conditioning formula gives the posterior predictive distribution of $y(\boldsymbol{x}_*)$, which is also Gaussian with the following predictive mean and variance:

$$\mathbb{E}[y(\boldsymbol{x}_*)] = \hat{\mu} + \boldsymbol{k}_N^\top\big[\hat{\sigma}^2\boldsymbol{I}_N + \boldsymbol{C}_N\big]^{-1}(\boldsymbol{y}_N - \hat{\mu}\mathbf{1}_N),$$
$$\text{Var}(y(\boldsymbol{x}_*)) = c(\boldsymbol{x}_*, \boldsymbol{x}_*; \hat{\theta}) - \boldsymbol{c}_N^\top\big[\hat{\sigma}^2\boldsymbol{I}_N + \boldsymbol{C}_N\big]^{-1}\boldsymbol{c}_N,$$

where $\boldsymbol{c}_N = [c(\boldsymbol{x}_i, \boldsymbol{x}_*; \hat{\theta}) : i = 1, \ldots, N]$ is a $N \times 1$ vector of the covariance values between the training data and the test data point.

The stationary GP model has many advantages such as modeling flexibility, analytical solution form and uncertainty quantification capability. Despite them, Gaussian Processes (GPs) scale poorly with data size, requiring $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ memory, which limits their applicability to moderately sized datasets. Moreover, in many practical settings, the underlying regression function is piecewise continuous and exhibits abrupt changes across unknown boundaries—behavior that stationary GPs are ill-equipped to model. Standard kernels impose global smoothness assumptions, leading to spurious correlations across discontinuities and biased estimates near regime shifts. Since stationary kernels depend solely on pairwise distances, they struggle to capture abrupt changes or heteroscedastic patterns. The Jump Gaussian Process (JGP) (Park 2022) addresses these limitations.

2.2 Jump Gaussian Processes (JGP)

JGP is best understood through the lens of local GP modeling (LAGP; Gramacy & Apley 2015). For each test location $\boldsymbol{x}_* \in \mathcal{X}$, a small subset of nearby training data is selected, $\mathcal{D}_n^{(*)} = \{(\boldsymbol{x}_i^{(*)}, y_i^{(*)})\}_{i=1}^n$, and a conventional stationary GP model is fitted to this local data. This approach is computationally efficient—$\mathcal{O}(n^3)$ versus $\mathcal{O}(N^3)$ when $n \ll N$—and can be massively parallelized across many test points (Gramacy et al. 2014). A key limitation of LAGP in estimating piecewise continuous surrogates is that local neighborhoods $\mathcal{D}_n^{(*)}$ may overlap partially or fully with discontinities. In such cases, LAGP can yield biased predictions (Park 2022) because the local data may mix training examples drawn from regions of the input space separated by abrupt regime shifts.

JGP addresses this issue by explicitly dividing the local data into two groups by regime shifts: data in the same regime as the test input $\boldsymbol{x}_*$ and the remainder. To accomplish this, JGP introduces a latent binary random variable $v_i^{(*)} \in \{0, 1\}$ indicating whether a training input $\boldsymbol{x}_i^{(*)}$ belongs to the same regime as $\boldsymbol{x}_*$ ($v_i^{(*)} = 1$) or not ($v_i^{(*)} = 0$). Conditional on $v_i^{(*)}$ values, $i = 1, \ldots, n$, the local data $\mathcal{D}_n^{(*)}$ is partitioned into two groups: $\mathcal{D}_* = \{i \in \{1, \ldots, n\} : Z_i^{(*)} = 1\}$ and $\mathcal{D}_o = \{1, \ldots, n\} \backslash \mathcal{D}_*$.

Only $\mathcal{D}_*$ contributes to predicting f at $\boldsymbol{x}_*$, while data in $\mathcal{D}_o$ are down-weighted via a uniform "outlier" likelihood. The full specification is completed by modeling $\mathcal{D}_*$ with a stationary

GP [Section 2.1], $\mathcal{D}_o$ with dummy likelihood $p(y_i^{(*)} \mid v_i^{(*)} = 0) \propto u$ for some constant, u, and assigning a prior to the latent variable $v_i^{(*)}$, via a sigmoid function π applied to a partitioning function $h(\boldsymbol{x}; \boldsymbol{\nu})$,

$$p(v_i^{(*)} = 1 | \boldsymbol{x}_i^{(*)}, \boldsymbol{\nu}) = \pi(h(\boldsymbol{x}_i^{(*)}; \boldsymbol{\nu})), \tag{2}$$

where $\boldsymbol{\nu}$ is another hyperparameter. The choice of the parametric partitioning function h determines the boundary separating $\mathcal{D}_o$ and $\mathcal{D}_*$. At the local level, linear or quadratic forms of h serves good Taylor approximations to complex domain boundaries around the local neighborhood of $\boldsymbol{x}_*$. For further details, see the original JGP paper (Park 2022). In this work, we adopt the linear form $h(\boldsymbol{x}; \boldsymbol{\nu}) = \boldsymbol{\nu}^T[1, \boldsymbol{x}]$.

Specifically, for $\boldsymbol{v}^{(*)} = (v_i^{(*)})_{i=1}^n$, $\boldsymbol{f}^{(*)} = (f(\boldsymbol{x}_i^{(*)}))_{i=1}^n$ and $\boldsymbol{\Theta} = \{\boldsymbol{\nu}, m^{(*)}, \theta^{(*)}, \sigma^2\}$, the JGP model is summarized as follows:

$$\begin{aligned}
p(\boldsymbol{y}_n \mid \boldsymbol{f}^{(*)}, \boldsymbol{v}^{(*)}, \boldsymbol{\Theta}) &= \prod_{i=1}^n \mathcal{N}_1(y_i^{(*)} | f_i^{(*)}, \sigma^2)^{v_i^{(*)}} u^{1-v_i^{(*)}}, \\
p(\boldsymbol{v}^{(*)} \mid \boldsymbol{\nu}) &= \prod_{i=1}^n \pi(h(\boldsymbol{x}_i^{(*)}; \boldsymbol{\nu}))^{v_i^{(*)}} (1 - \pi(h(\boldsymbol{x}_i^{(*)}; \boldsymbol{\nu})))^{1-v_i^{(*)}}, \\
p(\boldsymbol{f}^{(*)} \mid m^{(*)}, \theta^{(*)}) &= \mathcal{N}_n(\boldsymbol{f}^{(*)} \mid m^{(*)} \mathbf{1}_n, \boldsymbol{C}_n),
\end{aligned}$$

where $\boldsymbol{y}_n = (y_i^{(*)})_{i=1}^n$ and $\boldsymbol{C}_n$ is a $n \times n$ matrix with $c(x_i^{(*)}, x_j^{(*)}; \theta^{(*)})$ as its (i, j)th element. Parameters and latent indicators are learned via an EM-style algorithm, e.g., a variational EM variant (JGP-VEM) approximates the joint posterior over $\{\boldsymbol{v}^{(*)}, \boldsymbol{f}^{(*)}\}$, yielding similar predictive equations with uncertainty propagation.

Let $\hat{v}_i^{(*)}$ represent the MAP estimate of $v_i^{(*)}$ at the EM convergence and let $\hat{\mathcal{D}}_n^{(*)} = \{i : \hat{v}_i^{(*)} = 1\}$ denote the estimated in-regime subset and $\hat{\mathcal{D}}_o^{(*)} = \{1, \ldots, n\} \setminus \hat{\mathcal{D}}_n^{(*)}$ the out-of-regime subset. Let $\boldsymbol{y}_* = (y_i^{(*)}, i \in \hat{\mathcal{D}}_n^{(*)})$ and n_* denote the number of the elements in $\hat{\mathcal{D}}_n^{(*)}$. The posterior predictive mean and variance for $f(\boldsymbol{x}_*)$ are

$$\begin{aligned}
\mu_* &= \hat{m}^{(*)} + \boldsymbol{c}_{n,*}^\top \big(\hat{\sigma}^2 \boldsymbol{I}_{n_*} + \boldsymbol{C}_n^{(*)}\big)^{-1} \big(\boldsymbol{y}_* - m^{(*)} \mathbf{1}\big), \\
\sigma_*^2 &= c(\boldsymbol{x}_*, \boldsymbol{x}_*; \hat{\theta}^{(*)}) - \boldsymbol{c}_{n,*}^\top \big(\hat{\sigma}^2 \boldsymbol{I}_{n_*} + \boldsymbol{C}_n^{(*)}\big)^{-1} \boldsymbol{c}_{n,*},
\end{aligned} \tag{3}$$

where $\boldsymbol{c}_{n,*} = (c(\boldsymbol{x}_i^{(*)}, \boldsymbol{x}_*; \hat{\boldsymbol{\theta}}_*))_{i \in \hat{\mathcal{D}}_n^{(*)}}$ is a column vector of the covariance values between $\boldsymbol{y}_*$ and $f(\boldsymbol{x}_*)$, and $\boldsymbol{C}_n^{(*)} = (c(\boldsymbol{x}_i^{(*)}, \boldsymbol{x}_j^{(*)}; \hat{\boldsymbol{\theta}}_*))_{i,j \in \hat{\mathcal{D}}_n^{(*)}}$ is a square matrix of covariances evaluated for all pairs of $\boldsymbol{y}_*$. Here, $\hat{\sigma}^2$, $\hat{\theta}^{(*)}$ and $\hat{m}^{(*)}$ represent the MLEs of σ^2, $\boldsymbol{\theta}^{(*)}$ and $m^{(*)}$ respectively.

When the input dimension D is large, several challenges arise for JGP modeling. First, the number of hyperparameters grows quickly: a linear partition function $h(\boldsymbol{x}; \boldsymbol{\nu})$ requires $D + 1$ parameters, while a quadratic function demands on the order of $D^2 + 1$ parameters, quickly overwhelming the modest size of local neighborhoods. Second, there is a fundamental trade-off between bias and variance: enlarging the neighborhood yields more data for stable estimation of $\boldsymbol{\nu}$, but weakens the fidelity of the local Taylor approximation to complex boundaries; conversely, restricting to a small neighborhood preserves locality but risks overfitting due to limited data. Finally, the curse of dimensionality leads to sparse coverage in high-dimensional spaces, making it difficult to learn reliable regime boundaries without prior dimension reduction. These limitations motivate a unified framework that integrates dimensionality reduction directly into the JGP model, thereby enabling more effective modeling of high-dimensional, piecewise continuous functions.

2.3 Mahalanobis Gaussian Processes

Mahalanobis Gaussian Processes (AUEB & Lázaro-Gredilla 2013) extend traditional Gaussian process models by incorporating a built-in dimensionality reduction. The input vector $\boldsymbol{x}$ is linearly projected to $\boldsymbol{W}\boldsymbol{x}$ by a linear projection matrix $\boldsymbol{W} \in \mathbb{R}^{K \times D}$. The relation of the projected features to the response is modeled as a stationary GP model with the covariance kernel defined on the projected features. For instance, the squared exponential covariance (1) can be defined with the projected features as

$$K_W(\boldsymbol{x}_i, \boldsymbol{x}_j; \theta) = \sigma_f^2 \exp\left(-\frac{1}{2}(\boldsymbol{x}_i - \boldsymbol{x}_j)^\top \boldsymbol{W}^\top \boldsymbol{W}(\boldsymbol{x}_i - \boldsymbol{x}_j)\right). \quad (4)$$

To enable tractable learning, the authors introduced a variational inference framework for jointly estimating $\boldsymbol{W}$ along with the remaining GP hyperparameters.

Deep Variational Mahalanobis Gaussian Processes (DMGPs) (de Souza et al. 2022) extend MGPs by introducing a nonlinear, input-dependent projection. Unlike MGPs, which employ a single global linear projection $\boldsymbol{W}$, DMGP assigns each data point $\boldsymbol{x}$ its own projection matrix $\boldsymbol{W}(\boldsymbol{x})$. This results in a non-linear projection $g(\boldsymbol{x}) = \boldsymbol{W}(\boldsymbol{x})\boldsymbol{x}$. Each entry of $\boldsymbol{W}(\boldsymbol{x})$ is modeled as a function of $\boldsymbol{x}$, governed by a stationary GP. Within each row of $\boldsymbol{W}(\boldsymbol{x})$, the elements share a common GP prior with identical kernel hyperparameters. This construction enforces equal scaling of elements within a row through the shared kernel variance parameter, thereby achieving automatic relevance determination for the associated latent dimension.

DMGP assigns a distinct linear projection $\boldsymbol{W}(\boldsymbol{x})$ to each input location $\boldsymbol{x}$. However, this pointwise projection does not form a feasible combination with local models such as JGP, because the large number of the linear projection matrices easily makes an overfit to a small amount of local training data $\mathcal{D}_n^{(*)}$ in JGP. This motivates our main contribution, which integrates local projection into JGP under a variational framework. In the newly proposed DJGP, we seek for a locally constant approximation of $\boldsymbol{W}(\boldsymbol{x})$. When $\boldsymbol{W}(\boldsymbol{x})$ is a smooth function of $\boldsymbol{x}$, the local constant approximation can be justified by the zero-order Taylor approximation. Specifically, $\boldsymbol{W}(\boldsymbol{x})$ is approximately equal to $\boldsymbol{W}(\boldsymbol{x}_*)$ for the local data $\mathcal{D}_n^{(*)}$ nearby a test location $\boldsymbol{x}_*$. Under this formulation, the JGP model for $\boldsymbol{x}_*$ requires only a single projection matrix in addition to its standard parameters. This substantially reduces the number of parameters to estimate, improving feasibility while still capturing nonlinear projections through a piecewise constant structure.

3 Deep Jump Gaussian Process (DJGP)

Let $\mathcal{X}$ denote a domain of a function in $\mathbb{R}^D$. We consider a problem of estimating an unknown surrogate function which relates inputs $\boldsymbol{x} \in \mathcal{X}$ to a real response variable. We assume the existence of a nonlinear sufficient dimension reduction (for the unknown surrogate relation), which reduces the D-dimensional feature in $\mathcal{X}$ to a lower-dimensional feature in $\mathcal{Z} \subseteq \mathbb{R}^K$ via a continuously differentiable mapping,

$$g : \mathcal{X} \longrightarrow \mathcal{Z} \subseteq \mathbb{R}^K, \qquad K \ll D,$$

so that the response variable depends on $\boldsymbol{x}$ only through its reduced representation $\boldsymbol{z} = g(\boldsymbol{x})$. Therefore, we can introduce a reduced surrogate model f to relate $\boldsymbol{z}$ to the response variable. We assume f is assumed to be piecewise continuous in $\boldsymbol{z}$, so the composition function $f \circ g$ is also piecewise continuous in $\boldsymbol{x}$, given the assumed continuity of g. Specifically, there exists an (unknown) integer M and an unknown partition of $\mathcal{Z}$ into disjoint regions $\{\mathcal{Z}_m\}_{m=1}^M$ such that

$$f(\boldsymbol{z}) \;=\; \sum_{m=1}^{M} f_m(\boldsymbol{z})\, 1_{\mathcal{Z}_m}(\boldsymbol{z}), \tag{5}$$

where each local function f_m is a continuous function with its uncertainty modeled as a stationary Gaussian process (GP) with constant mean $\mu_m \in \mathbb{R}$ and a stationary covariance function $c_m(\cdot, \cdot)$. We assume mutual independence across regions:

$$\text{Independence:} \qquad f_m \text{ is independent of } f_\ell \text{ for } m \neq \ell. \tag{6}$$

which implies zero correlation between function values belonging to different regions.

For simplifying the model exposition, we restrict c_m to a parametric family $c(\cdot, \cdot; \theta) : \theta \in \Theta$, though the framework extends naturally to more general covariance functions. Let $\theta_m \in \Theta$ denote the region-specific covariance parameter so that $c_m(\cdot, \cdot) = c(\cdot, \cdot; \theta_m)$. We specifically consider the scale family, $c(\boldsymbol{z}, \boldsymbol{z}'; \theta_m) = a_m C(b_m ||\boldsymbol{z} - \boldsymbol{z}'||_2)$, where $||\boldsymbol{z} - \boldsymbol{z}'||_2$ is the Euclidean distance, $C(\cdot)$ is an isotropic correlation function with a unit length scale, $a_m > 0$ is the variance parameter, and $b_m > 0$ is the length scale parameter.

Finally, we assume heterogeneity in region means:

$$\text{Heterogeneity:} \qquad \mu_m \neq \mu_\ell, \theta_m \neq \theta_\ell \quad \text{for every pair of } m \neq \ell. \tag{7}$$

We aim to predict the surrogate response $f \circ g(\boldsymbol{x})$ at J test locations $\{\boldsymbol{x}^{(j)} \in \mathcal{X}, j = 1, \ldots, J\}$, given N noisy observations from the underlying model. Each observation at $\boldsymbol{x}_i$ is given as

$$y_i = f\big(g(\boldsymbol{x}_i)\big) + \epsilon_i, \qquad i = 1, \ldots, N, \tag{8}$$

where the noises $\epsilon_i \sim \mathcal{N}(0, \sigma^2(g(\boldsymbol{x}_i)))$ are independent. We denote the total training dataset $\mathcal{D}_{\boldsymbol{X}} = (\boldsymbol{X}, \boldsymbol{y}) = \{(\boldsymbol{x}_i, y_i), i = 1, \ldots, N\}$. The noise variance is assumed to change smoothly in the projected input $g(\boldsymbol{x}_i)$ and thus also smooth in the original input $\boldsymbol{x}_i$, so the variance is approximately constant around a small neighborhood of the projected input $g(\boldsymbol{x}_i)$.

3.1 Local Approximation

Modeling and estimating the complex functions g and f explicitly together with the unknown partition $\{\mathcal{Z}_m\}_{m=1}^M$ is challenging. Following the JGP framework, we instead seek a local approximation. For each test location $\boldsymbol{x}_*^{(j)}$, we first select a small subset of nearby training data—for example, the n nearest neighbors of $\boldsymbol{x}_*^{(j)}$ or a subset chosen by an existing local selection criterion (Gramacy & Apley 2015). We denote this local dataset by

$$\mathcal{D}_n^{(j)} = \{(\boldsymbol{x}_i^{(j)}, y_i^{(j)}) : i = 1, \ldots, n\}, \tag{9}$$

where $\boldsymbol{x}_i^{(j)}$ and $y_i^{(j)}$ represent the input vector and corresponding response of the the ith local data. For brevity, we introduce $\boldsymbol{X}^{(j)} = \{\boldsymbol{x}_i^{(j)}, i = 1, \ldots, n\}$ and $\boldsymbol{y}^{(j)} = \{y_i^{(j)}, i = 1, \ldots, n\}$.

Since we assumed the map g is assumed to be smooth and continuously differentiable, we apply a first-order Taylor approximation to g in a neighborhood of $\boldsymbol{x}_*^{(j)}$. For each local training point $(\boldsymbol{x}_i^{(j)}, y_i^{(j)})$ within the neighborhood,

$$g(\boldsymbol{x}_i^{(j)}) \approx g(\boldsymbol{x}_*^{(j)}) + \boldsymbol{W}_j(\boldsymbol{x}_i^{(j)} - \boldsymbol{x}_*^{(j)}). \tag{10}$$

The constant terms in the approximation do not affect the downstream GP modeling, so we omit the constant terms and define a local projection by $g(\boldsymbol{x}_i^{(j)}) \approx \boldsymbol{W}_j \boldsymbol{x}_i^{(j)}$.

The projection matrix $\boldsymbol{W}_j$ defines the direction of the local projection. To impose statistical correlation and encourage smooth variation of the projections over $\mathcal{X}$, we place a stationary Gaussian process prior on the collection of local projection matrices $\mathbf{W} = \{\mathbf{W}_j\}_{j=1}^J$. Specifically, let $w_{kd}^{(j)}$ denote the (k,d)-th entry of $\mathbf{W}_j$, and define the vector $\boldsymbol{w}_{kd} = [w_{kd}^{(1)}, \ldots, w_{kd}^{(J)}]$, which includes the (k,d)th entries across all local projection matrices. We then model $\boldsymbol{w}_{kd}$ as a Gaussian process with zero mean and the isotropic covariance function given by

$$c_{iso}(\boldsymbol{x}_*^{(j)}, \boldsymbol{x}_*^{(j')}; s, \ell_{w,k}) = s^2 \exp\left(- \frac{\|\boldsymbol{x}_*^{(j)} - \boldsymbol{x}_*^{(j')}\|^2}{2\ell_{w,k}^2} \right).$$

The squared exponential covariance models the correlation between local projection matrices, $\boldsymbol{W}j$ and $\boldsymbol{W}j'$, as a function of the distance between their corresponding test locations, $\boldsymbol{x}^{(j)}*$ and $\boldsymbol{x}^{(j')}*$. All entries share a common variance s^2, while elements within each row share a row-specific length-scale $\ell_{w,k}$, enforcing uniform scaling and enabling automatic relevance determination for each latent dimension. The joint prior is given by

$$p(\mathbf{W}|\boldsymbol{\Theta}_W) = \prod_{k=1}^{K} \prod_{d=1}^{D} \mathcal{N}\big(\boldsymbol{w}_{kd} | \mathbf{0}_J, \mathbf{C}_w^{(k)}\big), \tag{11}$$

where $\mathbf{0}_J$ is a J-dimensional column vector of zeros, and $\mathbf{C}_w^{(k)}$ is a $J \times J$ matrix with $c_{iso}(\boldsymbol{x}_*^{(j)}, \boldsymbol{x}_*^{(j')}; s, \ell_{w,k})$ as its (j, j') entry, and $\boldsymbol{\Theta}_W = (s, \ell_{w,1}, \ldots \ell_{w,K})$.

Conditioned on the local projection $\boldsymbol{W}_j$, the projected local dataset is defined as

$$\mathcal{D}_{\mathbf{W}_j,n}^{(j)} = \{(\boldsymbol{z}_i^{(j)}, y_i^{(j)}) : i = 1, \ldots, n, \boldsymbol{z}_i^{(j)} = \boldsymbol{W}_j \boldsymbol{x}_i^{(j)}\}. \tag{12}$$

By the mixture proposition in (5), these local data may originate from different regions, in which case the input–response relationship cannot be captured by a single Gaussian process. We follow JGP to model the mixture data. Specifically, in the jth local region, we introduce binary latent variables $v_i^{(j)} \in \{0, 1\}$, to indicate that the projected training input $\boldsymbol{z}_i^{(j)}$ belongs to the same region as the projected test point $\boldsymbol{W}_j\boldsymbol{x}_*^{(j)}$ ($v_i^{(j)} = 1$) or not ($v_i^{(j)} = 0$). Based on the indicator values, the local data $\mathcal{D}_{\boldsymbol{W}_j,n}^{(j)}$ can be partitioned into two groups: $\mathcal{D}_{\boldsymbol{W}_j,n}^{(j,1)} = \{i \in \{1,\dots,n\} : v_i^{(j)} = 1\}$ and the remainder $\mathcal{D}_{\boldsymbol{W}_j,n}^{(j,0)} = \{1,\dots,n\}\backslash\mathcal{D}_{\boldsymbol{W}_j,n}^{(j,1)}$. The first group belongs to the same region as the projected test location $\boldsymbol{W}_j\boldsymbol{x}_*^{(j)}$, so we use them to predict f at the test location. The second group is independent of f, based on the independence assumption (6), so it would be not used.

Since we are uncertain about the indicator values, we model them as random variables. We assign the prior probability to the indicator variables as in the JGP model (2),

$$p(v_i^{(j)} = 1|\boldsymbol{\nu}_j) = \pi(h(\boldsymbol{z}_i^{(j)}; \boldsymbol{\nu}_j)),$$

where $\pi(z) = 1/(1+e^{-z})$ is the sigmoid link function, and we use the linear decision function $h(\boldsymbol{z}_i^{(j)}; \boldsymbol{\nu}_j) = \boldsymbol{\nu}_j^T[1, \boldsymbol{z}_i^{(j)}]$. The logistic model divides the local data by the linear boundary, $\boldsymbol{\nu}_j^T[1, \boldsymbol{z}_i^{(j)}] = 0$. When the boundaries of the regions $\{\mathcal{Z}_m, m = 1, ..., M\}$ are smooth enough, the boundaries can be locally linearly approximated according to the Taylor approximation, so the use of the linear boundary to split the local data is justifiable. For more rough boundaries, one can use higher order polynomial functions.

The first group of the local data $\{(\boldsymbol{z}_i^{(j)}, y_i^{(j)}) : v_i^{(j)} = 1\}$ and the projected test point $\boldsymbol{W}_j\boldsymbol{x}_*^{(j)}$ belongs to the same region, denoted $m(j)$. Based on the model assumption (5), the input-output relation follows a stationary Gaussian process with the constant mean $\mu_{m(j)}$ and the covariance function $c(\cdot,\cdot;\theta_{m(j)})$. The region-specific covariance function is in the form of

$$c(\boldsymbol{z}, \boldsymbol{z}'; \theta_{m(j)}) = a_{m(j)}C(b_{m(j)}||\boldsymbol{z} - \boldsymbol{z}'||_2).$$

Since the local length scale parameter is redundant to the scale of the local projection matrix $\boldsymbol{W}_j$, we remove the length scale parameter. The removal makes the regional covariance function to have only the scale parameter $a_{m(j)}$ as

$$c(\boldsymbol{z}, \boldsymbol{z}'; a_{m(j)}) = a_{m(j)}C(||\boldsymbol{z} - \boldsymbol{z}'||_2).$$

Based on (8), y_i is a noisy realization of the Gaussian process

$$p\big(y_i^{(j)} \mid f_i^{(j)}, v_i^{(j)} = 1, \sigma_j^2\big) = \mathcal{N}\big(y_i^{(j)} \mid f_i^{(j)}, \sigma_j^2\big), \tag{13}$$

where σ_j^2 is a local constant approximation to $\sigma^2(g(\boldsymbol{x}))$ at $\boldsymbol{x}$ around $\boldsymbol{x}_*^{(j)}$. The Taylor approximation is justifiable given smoothness of $\sigma^2(\cdot)$ and $g(\cdot)$.

The other group of the local data $\{(\boldsymbol{z}_i^{(j)}, y_i^{(j)}) : v_i^{(j)} = 0\}$ is independent of the response variable at the projected test point $\boldsymbol{W}_j \boldsymbol{x}_*^{(j)}$. We treat them as outliers with respect to $f^{(j)}$ and assign them a uniform likelihood,

$$p\big(y_i^{(j)} \mid v_i^{(j)} = 0\big) = \frac{1}{u_j}. \tag{14}$$

Pulling all local data, latent variables and hyperparameters together, let $\boldsymbol{y}^{(j)} = (y_1^{(j)}, \ldots, y_n^{(j)})$, $\boldsymbol{v}^{(j)} = (v_1^{(j)}, \ldots, v_n^{(j)})$, $\boldsymbol{f}^{(j)} = (f_1^{(j)}, \ldots, f_n^{(j)})$ and $\boldsymbol{\Theta}^{(j)} = (\boldsymbol{\nu}_j, \sigma_j^2, \mu_{m(j)}, a_{m(j)})$. The conditional distribution is therefore

$$p\big(\boldsymbol{y}^{(j)} \big| \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)}, \boldsymbol{\Theta}^{(j)}\big) = \prod_{i=1}^{n} \Big[\mathcal{N}\big(y_i^{(j)} \mid f_i^{(j)}, \sigma_j^2\big)\Big]^{v_i^{(j)}} \left[\frac{1}{u_j}\right]^{1-v_i^{(j)}}, \text{ and}$$

$$p\big(\boldsymbol{v}^{(j)} \big| \boldsymbol{\Theta}^{(j)}\big) = \prod_{i=1}^{n} \Big[p(v_i^{(j)} = 1 | \boldsymbol{\nu}_j)\Big]^{v_i^{(j)}} \Big[1 - p(v_i^{(j)} = 1 | \boldsymbol{\nu}_j)\Big]^{1-v_i^{(j)}}.$$

The joint distribution for the local model is

$$p\big(\boldsymbol{y}^{(j)}, \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)} \big| \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}\big) = p\big(\boldsymbol{y}^{(j)} \big| \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)}, \boldsymbol{\Theta}^{(j)}\big) \times p\big(\boldsymbol{v}^{(j)} \big| \boldsymbol{\Theta}^{(j)}\big) \times p\big(\boldsymbol{f}^{(j)} \mid \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}\big),$$

where $p\big(\boldsymbol{f}^{(j)} \mid \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}\big) = \mathcal{N}(\boldsymbol{f}^{(j)} \mid \mu_{m(j)} \mathbf{1}_n, a_{m(j)} \boldsymbol{C}_{nn})$, and $\boldsymbol{C}_{nn}$ is a $n \times n$ matrix with $C(||\boldsymbol{z}_i^{(j)} - \boldsymbol{z}_{i'}^{(j)}||_2)$ as its (i, i')th element.

The full joint distribution is

$$p\big(\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{f}, \boldsymbol{W} \big| \boldsymbol{\Theta}\big) = p(\boldsymbol{W} | \boldsymbol{\Theta}_W) \times \prod_{j=1}^{J} p\big(\boldsymbol{y}^{(j)}, \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)} \big| \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}\big), \tag{15}$$

where $\boldsymbol{y} = (\boldsymbol{y}^{(1)}, \ldots, \boldsymbol{y}^{(J)})$, $\boldsymbol{v} = (\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(J)})$, $\boldsymbol{f} = (\boldsymbol{f}^{(1)}, \ldots, \boldsymbol{f}^{(J)})$, and $\boldsymbol{\Theta} = (\boldsymbol{\Theta}_W, \boldsymbol{\Theta}^{(1)}, \ldots, \boldsymbol{\Theta}^{(J)})$. Conditioned on the local projection matrix $\boldsymbol{W}_j$, the conditional model $p\big(\boldsymbol{y}^{(j)}, \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)} \big| \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}\big)$ is a local model, a JGP model specific to the local projection data $\mathcal{D}_{\boldsymbol{W}_j,n}^{(j)}$. The local project matrices are correlated through the global GP model $p(\boldsymbol{W})$. This unique two-layer GP/JGP model is referred to as the Deep JGP (DJGP) model.

3.2 Variational Inference

The statistical inference of the model parameters $\boldsymbol{\Theta}$ and the latent variables $\boldsymbol{f}$, $\boldsymbol{W}$, and $\boldsymbol{v}$ is analytically intractable due to the nonlinear dependencies introduced by the hierarchical structure. To address this challenge, we adopt a variational inference framework, following the sparse GP methodology introduced in MGP (AUEB & Lázaro-Gredilla 2013) and

DMGP (de Souza et al. 2022). Specifically, we introduce two sets of inducing variables: local inducing variables for the latent functions $\boldsymbol{f}$ to decouple the otherwise intractable dependencies between latent variables and hyperparameters, and global inducing variables for the projection process $\boldsymbol{W}$ to alleviate the prohibitive computational burden associated with repeated large-scale matrix inversions, when the number of the test locations is large.

Local inducing variables. For each local region associated with a test point $\boldsymbol{x}_*^{(j)}$, we introduce L_1 local inducing inputs $\tilde{\boldsymbol{z}}_\ell^{(j)} \in \mathbb{R}^K, \quad \ell = 1, \dots, L_1$, with corresponding inducing outputs $r_\ell^{(j)}$. These outputs are defined as standardized evaluations of the latent function $f_{m(j)}$, $r_\ell^{(j)} = (f_{m(j)}(\tilde{\boldsymbol{z}}_\ell^{(j)}) - \mu_{m(j)})/a_{m(j)}$, so that they are independent of the amplitude and mean hyperparameters $a_{m(j)}$ and $\mu_{m(j)}$, which improves identifiability. Collecting them as $\boldsymbol{r}^{(j)} = (r_\ell^{(j)})_{\ell=1}^{L_1} \in \mathbb{R}^{L_1}$, we assume the joint Gaussian prior $p(\boldsymbol{r}^{(j)}) = \mathcal{N}\Big(\boldsymbol{r}^{(j)} \mid \boldsymbol{0}_{L_1}, \boldsymbol{K}_r^{(j)}\Big)$, where $\boldsymbol{K}_r^{(j)} \in \mathbb{R}^{L_1 \times L_1}$ has entries $[\boldsymbol{K}_r^{(j)}]_{\ell\ell'} = C\Big(\|\tilde{\boldsymbol{z}}_\ell^{(j)} - \tilde{\boldsymbol{z}}_{\ell'}^{(j)}\|\Big)$. Conditioned on these inducing variables, the local latent function values $\boldsymbol{f}^{(j)}$ follow

$$p(\boldsymbol{f}^{(j)} \mid \boldsymbol{r}^{(j)}, \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}) = \mathcal{N}\Big(\boldsymbol{f}^{(j)} \mid \boldsymbol{K}_{fr}^{(j)}(\boldsymbol{K}_r^{(j)})^{-1}\boldsymbol{r}^{(j)},\ a_{m(j)}\boldsymbol{C}_{nn} - \boldsymbol{K}_{fr}^{(j)}(\boldsymbol{K}_r^{(j)})^{-1}(\boldsymbol{K}_{fr}^{(j)})^\top\Big),$$

where $\boldsymbol{K}_{fr}^{(j)} \in \mathbb{R}^{n \times L_1}$ with entries $[\boldsymbol{K}_{fr}^{(j)}]_{i\ell} = a_{m(j)}C(\|\boldsymbol{W}_j\boldsymbol{x}_i^{(j)} - \tilde{\boldsymbol{z}}_\ell^{(j)}\|^2)$, and $\boldsymbol{C}_{nn}$ denotes the kernel matrix constructed over the projected local inputs.

Global inducing variables. Similarly, for the projection process, we introduce L_2 global inducing inputs $\tilde{\boldsymbol{x}}_\ell \in \mathbb{R}^D, \quad \ell = 1, \dots, L_2$, with inducing outputs $\boldsymbol{R}_\ell \in \mathbb{R}^{K \times D}$, aggregated as $\boldsymbol{R} = (\boldsymbol{R}_\ell)_{\ell=1}^{L_2} \in \mathbb{R}^{L_2 \times K \times D}$. The inducing outputs are assumed to be drawn from the same GP as $\boldsymbol{W}_j$. Specifically, let $R_{\ell kd}$ denote the (k,d)-th entry of $\mathbf{R}_\ell$, and define the vector $\boldsymbol{R}_{:kd} = [R_{1kd}, \dots, R_{L_2kd}]$, which includes the (k,d)th entries across all inducing output matrices. Then, $\boldsymbol{R}_{:kd}$ is assumed to follow the same stationary GP as $\boldsymbol{\omega}_{kd}$. The prior distribution of $\boldsymbol{R}$ is given as

$$p(\boldsymbol{R}|\boldsymbol{\Theta}_W) = \prod_{k=1}^{K}\prod_{d=1}^{D}\mathcal{N}\Big(\boldsymbol{R}_{:kd} \,\Big|\, \boldsymbol{0}_{L_2}, \boldsymbol{K}_{\boldsymbol{R}}^{(k)}\Big), \tag{16}$$

where $\boldsymbol{K}_{\boldsymbol{R}}^{(k)}$ is a $L_2 \times L_2$ matrix with its (ℓ, ℓ')th entry as $c_{iso}(\tilde{\boldsymbol{x}}_\ell, \tilde{\boldsymbol{x}}_{\ell'}; s, \ell_{w,k})$. To reduce the computational burden when the number of test points J is large, we use the sparse Gaussian process approximation. It assumes that the conditional independence of the elements in

$\boldsymbol{w}_{k,d}$ conditioned on $\boldsymbol{R}_{:kd}$, which would give the conditional distribution as

$$p(\boldsymbol{W} \mid \boldsymbol{R}, \boldsymbol{\Theta}_W) = \prod_{k=1}^{K} \prod_{d=1}^{D} \mathcal{N}\Big(\boldsymbol{w}_{kd} \;\Big|\; \boldsymbol{K}_{\boldsymbol{WR}}^{(k)} (\boldsymbol{K}_R^{(k)})^{-1} \boldsymbol{R}_{:kd},\ \boldsymbol{\Lambda}_{\boldsymbol{W}}^{(k)} - \boldsymbol{K}_{\boldsymbol{WR}}^{(k)} (\boldsymbol{K}_{\boldsymbol{R}}^{(k)})^{-1} \boldsymbol{K}_{\boldsymbol{WR}}^{(k)\top}\Big).$$

where $\boldsymbol{\Lambda}_{\boldsymbol{W}}^{(k)}$ is a $J \times J$ diagonal matrix with its jth diagonal element equal to $c_{iso}(\boldsymbol{x}_*^{(j)}, \boldsymbol{x}_*^{(j)}; s, \ell_{w,k})$, $\boldsymbol{K}_{\boldsymbol{WR}}^{(k)}$ is a $J \times L_2$ matrix with its (j, ℓ)th element equal to $c_{iso}(\boldsymbol{x}_*^{(j)}, \tilde{\boldsymbol{x}}_\ell; s, \ell_{w,k})$.

Variational family and ELBO. The full posterior distribution with the two sets of the inducing variables is

$$\begin{aligned} &p\big(\boldsymbol{v}, \boldsymbol{f}, \boldsymbol{W}, \boldsymbol{R}, \boldsymbol{r} \big| \boldsymbol{y}, \boldsymbol{\Theta}\big) \propto p(\boldsymbol{W}|\boldsymbol{R}, \boldsymbol{\Theta}_W) \times p(\boldsymbol{R}|\boldsymbol{\Theta}_W) \\ &\quad \times \prod_{j=1}^{J} p\big(\boldsymbol{y}^{(j)} \big| \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)}, \boldsymbol{\Theta}^{(j)}\big) \times p\big(\boldsymbol{v}^{(j)} \big| \boldsymbol{\Theta}^{(j)}, \boldsymbol{W}_j\big) \times p(\boldsymbol{f}^{(j)} \mid \boldsymbol{r}^{(j)}, \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)}) \times p(\boldsymbol{r}^{(j)}) \end{aligned} \tag{17}$$

We approximate it with the following factorized variational distribution:

$$q\big(\boldsymbol{v}, \boldsymbol{f}, \boldsymbol{W}, \boldsymbol{R}, \boldsymbol{r}\big) = p\big(\boldsymbol{W} \mid \boldsymbol{R}, \boldsymbol{\Theta}_W\big)\, q\big(\boldsymbol{R}\big) \prod_{j=1}^{J} \Big[q(\boldsymbol{v}^{(j)})\, p(\boldsymbol{f}^{(j)} \mid \boldsymbol{r}^{(j)}, \boldsymbol{W}_j, \boldsymbol{\Theta}^{(j)})\, q(\boldsymbol{r}^{(j)}) \Big]. \tag{18}$$

where

$$\begin{aligned} q\big(\boldsymbol{v}^{(j)}\big) &= \prod_{i \in \mathcal{D}_n^{(j)}} q\big(v_i^{(j)}\big) \qquad \text{with } q\big(v_i^{(j)}\big) = \mathrm{Bernoulli}\big(\rho_i^{(j)}\big) \\ q\big(\boldsymbol{r}^{(j)}\big) &= \mathcal{N}\big(\boldsymbol{\mu}_r^{(j)}, \boldsymbol{\Sigma}_r^{(j)}\big), \\ q(\boldsymbol{R}) &= \prod_{\ell=1}^{L_2} \prod_{k=1}^{K} \prod_{d=1}^{D} q\big(R_{\ell kd}\big) = \prod_{k,d} q(\boldsymbol{R}_{:kd}) \qquad \text{with } q\big(R_{\ell kd}\big) = \mathcal{N}\big(\mu_{lkd}, \sigma_{lkd}^2\big). \end{aligned} \tag{19}$$

Here, $\boldsymbol{R}_{:kd} := (R_{1kd}, \ldots, R_{L_2kd})^\top \in \mathbb{R}^{L_2}$ denotes the slice of $\boldsymbol{R}$ along the inducing-point index ℓ for fixed (k, d). Accordingly, under (19) we have $q(\boldsymbol{R}_{:kd}) = \mathcal{N}(\boldsymbol{\mu}_{kd}, \boldsymbol{\Sigma}_{kd})$ with $\boldsymbol{\mu}_{kd} = (\mu_{1kd}, \ldots, \mu_{L_2kd})^\top$ and $\boldsymbol{\Sigma}_{kd} = \mathrm{diag}(\sigma_{1kd}^2, \ldots, \sigma_{L_2kd}^2)$.

The distribution parameters, $\rho_i^{(j)}$, $\boldsymbol{\mu}_r^{(j)}, \boldsymbol{\Sigma}_r^{(j)}$, μ_{lkd} and σ_{lkd}, are unknown, variational parameters to optimize. We denote them collectively by $\boldsymbol{\Theta}_V$.

Under the variational family above, the evidence lower bound (ELBO) can be written as

$$\begin{aligned} \mathcal{L} = \sum_{j=1}^{J} \Big\{ &\mathbb{E}_{q(\boldsymbol{r}^{(j)})\, q(\boldsymbol{W}_j)\, q(\boldsymbol{v}^{(j)})} \Big[\log p\big(\boldsymbol{y}^{(j)} \mid \boldsymbol{v}^{(j)}, \boldsymbol{f}^{(j)}, \boldsymbol{\Theta}^{(j)}\big) \Big] \\ &+ \mathbb{E}_{q(\boldsymbol{W}_j)\, q(\boldsymbol{v}^{(j)})} \Big[\log p\big(\boldsymbol{v}^{(j)} \mid \boldsymbol{\Theta}^{(j)}, \boldsymbol{W}_j\big) - \log q(\boldsymbol{v}^{(j)}) \Big] - \mathrm{KL}\big(q(\boldsymbol{r}^{(j)}) \,\|\, p(\boldsymbol{r}^{(j)})\big) \Big\} \\ - \ &\mathrm{KL}\big(q(\boldsymbol{R}) \,\|\, p(\boldsymbol{R} \mid \boldsymbol{\Theta}_W)\big), \end{aligned} \tag{20}$$

where $q(\boldsymbol{W}_j) = \mathbb{E}_{q(\boldsymbol{R})}[\, p(\boldsymbol{W}_j \mid \boldsymbol{R})\,]$.

To enable efficient gradient-based optimization, we derive a computable closed-form for (20). We first define the expected kernel statistics

$$\Psi_1^{(j)} := \mathbb{E}_{q(\boldsymbol{W}_j)}[\boldsymbol{K}_{fr}^{(j)}], \qquad \Psi_2^{(i,j)} := \mathbb{E}_{q(\boldsymbol{W}_j)}[\boldsymbol{K}_{rf}^{(i,j)} \boldsymbol{K}_{fr}^{(i,j)}]$$

where we denote $\boldsymbol{K}_{fr}^{(i,j)} \in \mathbb{R}^{1\times L_1}$ for the i-th row of $\boldsymbol{K}_{fr}^{(j)}$, , and accordingly $\boldsymbol{K}_{rf}^{(i,j)} \triangleq (\boldsymbol{K}_{fr}^{(i,j)})^\top \in \mathbb{R}^{L_1\times 1}$. For each local region j and training neighbor i, we define the following auxiliary scalars to represent the uncertainty propagation through the latent layers:

$$Q_{j,i} := \frac{(y_i^{(j)})^2 - 2y_i^{(j)}\zeta_{j,i} + A_{j,i} + B_{j,i}}{2\sigma_j^2},$$

where $\zeta_{j,i}$ denotes the ith element of $\Psi_1^{(j)}(\boldsymbol{K}_r^{(j)})^{-1}\boldsymbol{\mu}_r^{(j)}$, $A_{j,i} := a_{m(j)} - \mathrm{tr}\Big((\boldsymbol{K}_r^{(j)})^{-1}\Psi_2^{(i,j)}\Big)$, and $B_{j,i} := \mathrm{tr}\Big((\boldsymbol{K}_r^{(j)})^{-1}\Psi_2^{(i,j)}(\boldsymbol{K}_r^{(j)})^{-1}\big(\boldsymbol{\mu}_r^{(j)}\boldsymbol{\mu}_r^{(j)\top} + \boldsymbol{\Sigma}_r^{(j)}\big)\Big)$.

Combining these with the expected log-likelihood of the local gating function, we define the local log-evidence components $S_1^{i,j}$ and $S_2^{i,j}$:

$$\begin{aligned}
S_1^{i,j} &:= -\tfrac{1}{2}\log(2\pi\sigma_j^2) - Q_{j,i} + \mathbb{E}_{q(\boldsymbol{W}_j)}\log\sigma\big(\boldsymbol{\nu}_j^\top[1, \boldsymbol{W}_j\boldsymbol{x}_i^{(j)}]\big),\\
S_2^{i,j} &:= -\log u_j + \mathbb{E}_{q(\boldsymbol{W}_j)}\log\big(1-\sigma(\boldsymbol{\nu}_j^\top[1, \boldsymbol{W}_j\boldsymbol{x}_i^{(j)}])\big).
\end{aligned}$$

Plugging these into (20), the ELBO admits the following final computable form:

$$\begin{aligned}
\mathcal{L} =& \sum_{j=1}^{J}\sum_{i\in\mathcal{D}_n^{(j)}} \log\Big(\exp(S_1^{i,j}) + \exp(S_2^{i,j})\Big)\\
&+ \sum_{k=1}^{K}\sum_{d=1}^{D}\frac{1}{2}\Big[\log\frac{|\boldsymbol{K}_R^{(k)}|}{|\boldsymbol{\Sigma}_{kd}|} - L_2 + \mathrm{tr}\big((\boldsymbol{K}_R^{(k)})^{-1}\boldsymbol{\Sigma}_{kd}\big) + \boldsymbol{\mu}_{kd}^\top(\boldsymbol{K}_R^{(k)})^{-1}\boldsymbol{\mu}_{kd}\Big],
\end{aligned} \tag{21}$$

where $K_R^{(k)}$ and (μ_{kd}, Σ_{kd}) are defined in equation (16) and (19) respectively. The detailed derivation are provided in Appendix A.

In practice, we maximize $\mathcal{L}$ in (21) by stochastic gradient ascent with respect to the variational parameters $\boldsymbol{\Theta}_V$, the inducing inputs $\tilde{\boldsymbol{x}} = (\tilde{\boldsymbol{x}}_\ell)_{\ell=1}^{L_2}$, and the model hyperparameters $\boldsymbol{\Theta}$. We fixed the local inducing inputs $\{\tilde{\boldsymbol{z}}_\ell^{(j)}\}_{\ell=1}^{L_1}$ to randomly sampled values, specifically, $\tilde{\boldsymbol{z}}_\ell^{(j)} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_Q)$, because the learning output was not very sensitive to the choices. In contrast, the global inducing inputs $\{\tilde{\boldsymbol{x}}_\ell\}_{\ell=1}^{L_2}$ are treated as learnable parameters and jointly optimized with the other model parameters.

To encourage a reasonable initialization before optimization, we draw them around the empirical mean of the training inputs with random perturbations proportional to the empirical standard deviation, that is, $\tilde{\boldsymbol{x}}_\ell = \bar{\boldsymbol{x}} + \boldsymbol{\epsilon}_\ell \odot \boldsymbol{\sigma}_x$ with $\boldsymbol{\epsilon}_\ell \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_D)$, where $\bar{\boldsymbol{x}}$ and $\boldsymbol{\sigma}_x$ denote the element-wise mean and standard deviation of the training data. This scheme ensures that the global inducing points are well spread within the data manifold and provides a stable starting point for subsequent variational optimization.

Following (AUEB & Lázaro-Gredilla 2013), the global kernel hyperparameters and inducing inputs are optimized via a Type-II maximum likelihood (empirical Bayes) approach, which has been shown to yield robust and computationally efficient performance for Mahalanobis-type Gaussian process models.

3.3 Prediction

To mitigate any suboptimality of the variational approximation, we use a two-stage, sampling-based prediction. We follow the approach of (AUEB & Lázaro-Gredilla 2013) (2013, Sec. 2.5, Eqs. (17)–(18)) and adapt it to our model. For each test location, we takes a random sample of the projection matrices drawn from its estimated variational posterior distribution $q(\boldsymbol{W}_j)$. Each sampled projection defines a different low-dimensional embedding, and the predictive mean and variance are obtained by averaging the corresponding GP predictions. Sampling from $q(\boldsymbol{W}_j)$ provides a computationally efficient and statistically robust way to propagate the uncertainty of the learned projection into the Jump GP predictions.

Specifically, for each test input $\boldsymbol{x}^{(j)}$, we first draw M_c samples $\boldsymbol{W}_j^{(m)} \sim q(\boldsymbol{W}_j), \quad m = 1, \dots, M_c$, from the variational posterior over the projection matrix. Each sample defines a low-dimensional embedding of the training data $\boldsymbol{z}_i^{(j,m)} = \boldsymbol{W}_j^{(m)} \boldsymbol{x}_i^{(j)}$ for $i = 1, \dots, n$. Let $\boldsymbol{Z}^{(i,m)}$ denote the n locally embedded training inputs. We then fit a standard Jump GP on $\big(\boldsymbol{Z}^{(i,m)}, \boldsymbol{y}^{(j)}\big)$ to obtain the predictive mean $\mu_*^{(j,m)}$ and variance $\sigma_*^{2\,(j,m)}$, according to the posterior expressions in Eq. (3) of Section 2.2). The final prediction for region j is computed by averaging over the M_c Monte Carlo samples. Finally, we aggregate via Monte Carlo:

$$\mu_*^{(j)} = \frac{1}{M_c} \sum_{m=1}^{M_c} \mu_*^{(j,m)}, \qquad \sigma_*^{2\,(j)} = \frac{1}{M_c} \sum_{m=1}^{M_c} \Big[\sigma_*^{2\,(j,m)} + \big(\mu_*^{(j,m)} - \mu_*^{(j)}\big)^2\Big].$$

This procedure leverages the global consistency of the variational posterior for each $\boldsymbol{W}_j$ while retaining the local adaptivity and uncertainty calibration of Jump GP in the learned subspace. The detailed pseudo-algorithm could be found in Algorithm 1.

Algorithm 1 DJGP: Variational Training & Prediction

Require: Region data $\{(\mathcal{D}_n^{(j)}, \boldsymbol{x}_*^{(j)})\}_{j=1}^J$, initial values of the variational parameters $\boldsymbol{\Theta}_V$, inducing inputs/outputs $\{(\tilde{z}_\ell^{(j)}, r_\ell^{(j)}) : \ell = 1, ..., L_1, j = 1, \ldots, J\}$ and $\{(\tilde{x}_\ell, \boldsymbol{R}_\ell) : \ell = 1, ..., L_2\}$, and other hyperparameters $\boldsymbol{\Theta}$, learning rate η, max iterations S, M_c

Ensure: Posterior approximations and predictive distribution at test points

for $s = 1$ to S **do**

  Compute $\mathcal{L}$ in (21) and its gradients w.r.t. all variational and model parameters $(\boldsymbol{\Theta}_V, \tilde{\boldsymbol{x}}, \boldsymbol{\Theta})$

  Update parameters by gradient ascent: $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{L}$

  Enforce positivity constraints on variances and lengthscales

  If ELBO has converged **then break**

end for

Prediction:

for each test region j **do**

  Compute posterior $q(\boldsymbol{W}_j)$ from $q(\boldsymbol{R})$ and conditional GP prior

  Draw M_c samples $\{\boldsymbol{W}_j^{(m)}\} \sim q(\boldsymbol{W}_j)$

  for $m = 1$ to M_c **do**

    Project data: $\tilde{\boldsymbol{Z}}^{(j)} = \boldsymbol{W}_j^{(m)} \boldsymbol{X}^{(j)}$, $\tilde{\boldsymbol{z}}_*^{(j)} = \boldsymbol{W}_j^{(m)} \boldsymbol{x}_*^{(j)}$

    Fit local Jump GP on $(\tilde{\boldsymbol{Z}}^{(j)}, \boldsymbol{y}^{(j)})$ to predict $\mu_*^{(j,m)}, \sigma_*^{2(j,m)}$

  end for

  Aggregate $\mu_*^{(j)} = \frac{1}{M_c} \sum_m \mu_*^{(j,m)}$, $\sigma_*^{2(j)} = \frac{1}{M_c} \sum_m \left[\sigma_*^{2(j,m)} + (\mu_*^{(j,m)} - \mu_*^{(j)})^2\right]$

end for

return $\{\mu_*^{(j)}, \sigma_*^{2(j)}\}_{j=1}^J$

---

4 Theoretical Results

This section presents the prediction error for DJGP. The prediction error can be decomposed into four sources: (i) local classification errors (errors classifying errors), (ii) projection estimation error, (iii) local linearization error, and (iv) latent-space GP regression error.

This decomposition elucidates how the core structural components of DJGP contribute to the overall prediction risk. Detailed proofs are provided in Appendix B.

We start by introducing some useful notations to develop the theoretical result. Given a random test point $\boldsymbol{x}_*$ drawn from an unknown input distribution $P_{\mathcal{X}}$ on $\mathcal{X}$, let $\mathcal{D}_n^{(*)} \subseteq \{1, \ldots, n\}$ denote the index set of the n nearest neighbors of $\boldsymbol{x}_*$ from the full training set $\mathcal{D}_X$, and define the neighborhood radius by $\rho_r(\boldsymbol{x}_*) := \max_{i \in \mathcal{D}_n^{(*)}} \|\boldsymbol{x}_i - \boldsymbol{x}_*\|$. Let $(\boldsymbol{W}_*, b_*)$ denote the first-order local linear approximation of $g(\boldsymbol{x})$ at $\boldsymbol{x} = \boldsymbol{x}_*$, satisfying

$$g(\boldsymbol{x}_*) = \boldsymbol{W}_* \boldsymbol{x}_* + b_*. \tag{22}$$

Without loss of generality, we set $b_* = 0$. Let $\boldsymbol{W} \in \mathbb{R}^{K \times D}$ denote the fitted projection matrix, and let $\hat{f}_X^{(\boldsymbol{W})}$ denote the JGP predictive mean at $\boldsymbol{x}_*$ based on the local projected data $\{(\boldsymbol{z}_i^{(\boldsymbol{W})} = \boldsymbol{W}\boldsymbol{x}_i,\ y_i = f(g(\boldsymbol{x}_i)) + \epsilon_i) : i \in \mathcal{D}_n^{(*)}\}$.

Our goal is to bound the prediction risk $\mathcal{R} := \mathbb{E}\big[(\hat{f}_X^{(\boldsymbol{W})} - f(g(\boldsymbol{x}_*)))^2\big]$, where the expectation is taken over both $\boldsymbol{x}_*$ and the training dataset $\mathcal{D}_X = (\boldsymbol{X}, \boldsymbol{y})$.

We next introduce the assumptions used in the analysis.

Assumption 1 (Smooth latent map)**.** *The function g is twice continuously differentiable in a neighborhood of x with bounded Hessian: there exists $M_g \geq 0$ such that*

$$\|\nabla^2 g(x)\| \leq M_g.$$

Let $\{\mathcal{Z}_m\}_{m=1}^M$ be a partition of the latent space $\mathcal{Z}$ into regions, and define

$$\partial\mathcal{Z} := \bigcup_{m \neq m'} \left(\partial\mathcal{Z}_m \cap \partial\mathcal{Z}_{m'}\right)$$

as the union of region boundaries. For each region m, define the within-region function

$$f_m := f|_{\mathcal{Z}_m}, \qquad \text{i.e.,} \qquad f_m(z) = f(z) \text{ for all } z \in \mathcal{Z}_m.$$

For a boundary point $z \in \partial\mathcal{Z}_m \cap \partial\mathcal{Z}_{m'}$, let z^+ and z^- denote the points approaching z from $\mathcal{Z}_m$ and $\mathcal{Z}_{m'}$, respectively, along the normal direction to the boundary. Define the jump magnitude across region boundaries as $\Delta_f := \max_{m \neq m'} \sup_{z \in \partial\mathcal{Z}_m \cap \partial\mathcal{Z}_{m'}} \big|f_m(z^+) - f_{m'}(z^-)\big|$.

Assumption 2 (Within-region regularity of f)**.** *For each region m, assume $f_m \in \mathcal{H}_{c_m}$, where $\mathcal{H}_{c_m}$ is the RKHS induced by a positive definite and locally Lipschitz kernel c_m, and there exists $B_f \geq 0$ such that*

$$\|f_m\|_{\mathcal{H}_{c_m}} \leq B_f.$$

Let $Z \in \mathcal{Z}$ denote a latent input, and let $r(Z) \in \{0,1\}$ be the ground-truth region label for the gating task under consideration.[1] Define the posterior class probability and the plug-in classifier by

$$\eta(z) := \Pr(r(Z) = 1 \mid Z = z), \qquad \hat{r}(z) := \mathbb{I}\{\hat{\eta}(z) \geq 1/2\},$$

where $\hat{\eta}(z)$ is an estimator of $\eta(z)$ trained on n gating samples. The gating regression error is defined as

$$\epsilon_n := \mathbb{E}_Z\big[|\hat{\eta}(Z) - \eta(Z)|\big], \tag{23}$$

where the expectation is taken over $Z \sim P_{\mathcal{Z}}$, the latent distribution induced by $P_{\mathcal{X}}$.

Assumption 3 (Tsybakov margin condition (Tsybakov 2004))**.** *There exist constants $C_0 > 0$ and $\alpha > 0$ such that, for all $t > 0$,*

$$\Pr\big(|\eta(Z) - 1/2| \leq t\big) \leq C_0\, t^{\alpha}.$$

Under Assumption 3, the plug-in classifier satisfies the standard bound

$$\Pr\big(\hat{r}(Z) \neq r(Z)\big) \lesssim \epsilon_n^{1+\alpha}$$

(e.g., Audibert & Tsybakov 2007, Tsybakov 2004). Thus, ϵ_n governs the accuracy of the learned gating boundary. For a well-specified parametric gate, typically $\epsilon_n = O(n^{-1/2})$, which yields

$$\Pr\big(\hat{r}(Z) \neq r(Z)\big) \lesssim n^{-(1+\alpha)/2}.$$

We now state the main risk bound for DJGP. The bound clarifies how DJGP balances dimensionality reduction, gating accuracy, and local approximation.

Theorem 1 (Overall Risk Bound for DJGP)**.** *Under Assumptions 1–3,*

$$\begin{aligned}\mathcal{R} \lesssim\ & K D\, L_2^{-1} + \mathrm{KL}\big(q(R) \,\|\, p(R)\big) + \|\mathbb{E}_q \boldsymbol{W} - \boldsymbol{W}_*\|_F^2 + \mathbb{E}[\rho_r(\boldsymbol{X})^4] + \sigma^2 \\ & + \big(\tau^2 + \tau^{-1}\epsilon_n\big)\Delta_f^2 + \mathrm{GP}_{\mathrm{oracle}}(n, K),\end{aligned} \tag{24}$$

where $\tau \in (0,1)$ and

$$\mathrm{GP}_{\mathrm{oracle}}(n,K) \lesssim \begin{cases} B_f^2\, \dfrac{(\log n)^{K+1}}{n}, & \textit{for the squared exponential kernel,} \\[2ex] B_f^2\, n^{-2\nu_M/(2\nu_M+K)}, & \textit{for the Matérn}(\nu_M) \textit{ kernel,} \end{cases}$$

[1] The analysis is stated for a binary gate, which is the standard setting for Tsybakov's margin condition. In the multi-region case, the same argument applies to an appropriate one-vs-one or one-vs-rest gate associated with a local boundary.

with $\nu_M > 0$ the Matérn smoothness parameter.

The term $O(KD\,L_2^{-1})$ is the Nyström approximation error and vanishes as the number of global inducing points L_2 increases. The term $\mathbb{E}[\rho_r(\boldsymbol{X})^4]$ is the local linearization error from approximating $g(\cdot)$ by its first-order Taylor expansion. The gating contribution is controlled by $(\tau^2 + \tau^{-1}\epsilon_n)\Delta_f^2$; choosing $\tau \asymp \epsilon_n^{1/3}$ yields the rate $O(\Delta_f^2\epsilon_n^{2/3})$. When the projection is estimated accurately and local neighborhoods are sufficiently dense, the dominant term becomes $\mathrm{GP}_{\mathrm{oracle}}(n, K)$, which depends on the intrinsic dimension K rather than the ambient dimension D. This shows that DJGP adapts to the latent low-dimensional geometry while remaining robust to jump discontinuities.

5 Synthetic Dataset Experiments

We assess the performance of DJGP on synthetic datasets, comparing against several baseline methods[2]. We report both root mean squared error (RMSE) and continuous ranked probability score (CRPS) as evaluation metrics; we favor CRPS over negative log predictive density (NLPD) because of NLPD's sensitivity to outliers. When it compares a Gaussian predictive distribution $\mathcal{N}(\mu_j, \sigma_j^2)$ at the jth test site ($j = 1, ..., J$) with the test response $y_*^{(j)}$, the two metrics are defined as RMSE $= 1/J\sqrt{\sum_{j=1}^{J}(y_*^{(j)} - \mu_j)^2}$ and CRPS $= 1/J\sum_{j=1}^{J}\sigma_j\Big[\,\tilde{y}_j\big(2\Phi(\tilde{y}_j) - 1\big) + 2\phi(\tilde{y}_j) - 1/\sqrt{\pi}\Big]$, with $\tilde{y}_j = (y_*^{(j)} - \mu_j)/\sigma_j$, and $\Phi(\cdot)$ and $\phi(\cdot)$ the standard normal CDF and PDF, respectively.

Benchmark Methods The first baseline is the Jump Gaussian Process (**JGP**) proposed by Park et al. (Park 2022), implemented with classification EM and linear partition boundaries. We restrict JGP to linear separators to avoid overfitting in high-dimensional settings and to ensure a fair comparison with DJGP. The second baseline is **JGP-SIR**, which applies sliced inverse regression (SIR) (Li 1991) to reduce the input dimension before fitting a standard JGP on the projected features. The third baseline is **JGP-AE**, which employs an autoencoder for dimensionality reduction, followed by fitting JGP in the learned low-dimensional space. The autoencoder is implemented as a multi-layer perceptron (MLP) for both encoder and decoder. Fourth, we include a two-layer, doubly-stochastic Deep

[2]The complete Python codebase—including scripts for benchmark models—is available at `https://github.com/crushonyfg/DJGP`.

Gaussian Process (**DGP**) (Gardner et al. 2018). This model can also be interpreted as a Gaussian Process Latent Variable Model (GP-LVM). We omit the Deep Mahalanobis Gaussian Process (DMGP), as it performs comparable to **DGP** (de Souza et al. 2022).

Synthetic Datasets Construction Synthetic datasets are generated using a two-stage framework. In the first stage, we construct a latent feature space $\mathcal{Z}$ of dimension K and define the response relationship as $y = f(\boldsymbol{z})+\epsilon$. In the second stage, we apply dimensionality expansion by simulating the inverse of a smooth projection $g^{-1} : \mathcal{Z} \to \mathcal{X}$, lifting the latent dimension K to an observed dimension D. The resulting datasets therefore have D-dimensional inputs.

The first stage generates $\{(\boldsymbol{z}, y), y = f(\boldsymbol{z}) + \epsilon\}$, using the two following scenarios:

- **L2 Dataset** $K = 2$**:** A two-dimensional synthetic dataset defined on $[-0.5, 0.5]^2$, partitioned into multiple regions. Within each region m, the response surface is drawn from an independent GP with mean $\mu_m \in \{0, 27\}$ and squared-exponential covariance $c(\boldsymbol{z}, \boldsymbol{z}'; \theta_m) = \theta_{m1} \exp\left\{-\frac{1}{\theta_{m2}}(\boldsymbol{z} - \boldsymbol{z}')^\top(\boldsymbol{z} - \boldsymbol{z}')\right\}$, where $\theta_{m1} = 9$ and $\theta_{m2} = 200$. The large length-scale yields nearly constant surfaces within each region, effectively emulating piecewise-constant responses with Gaussian noise ($\sigma^2 = 4$). A total of 1,100 points are sampled uniformly, with 1,000 used for training and 100 for testing.

- **LH Dataset** $K = 2, 3, 4$**:** A higher-dimensional dataset defined on $[-0.5, 0.5]^K$ for $K = 2, 3, 4$. We define $K + 1$ partitioning functions:

  $$f_0(\boldsymbol{z}) = \sum_{i=1}^{K} z_i^2 - 0.4^2, \quad f_j(\boldsymbol{z}) = \sum_{i=1}^{K} z_i^2 - z_j^2 + (z_j + r_j \cdot 0.5)^2 - 0.3^2, \quad j = 1, \ldots, K,$$

  where $r_j \sim \text{Uniform}\{\pm 1\}$. Each f_j partitions the domain into two regions, and the region index is defined as $\text{region}(\boldsymbol{z}) = \sum_{j=0}^{K} 2^j \, \mathbb{I}_{\mathcal{Z}_{j,+}}(\boldsymbol{z})$. Responses are sampled from region-dependent GPs with mean $\mu_m \sim \text{Uniform}\{-13.5m,\ 13.5m\}$ and the same covariance as above. Gaussian noise $\epsilon \sim \mathcal{N}(0, 4)$ is added. We generate N training samples uniformly and 100 test points concentrated near region boundaries.

To emulate high-dimensional inputs, the second stage lifts $\boldsymbol{z} \in \mathbb{R}^K$ to high dimensional inputs $\boldsymbol{x} \in \mathbb{R}^D$: random projection (RP), random Fourier features (RF), polynomial expansion (PE), and autoencoder-based expansion (AE). Details are provided in Appendix C.

Each dataset is thus defined by a combination of a first-stage choice (four choices: L2 or LH with $K = 2, 3, 4$) and a second-stage choice (four schemes). In total, we have 16 synthetic datasets generated.

Implementation Details For each dataset setting, we repeat the experiment 10 times to account for randomness and report averaged results. Hyperparameters are tuned via five-fold cross-validation on the training set for the first run of each setting; the selected values are then reused in subsequent runs to reduce computational cost. Specifically, the latent dimension Q is chosen from $\{3, 5, 7\}$, and the numbers of inducing points (L_1, L_2) are selected from the grid $\{2, 4, 6\} \times \{20, 40, 60\}$. After selection, each model is retrained on the full training set and evaluated on a held-out test set.

To keep the search space manageable, we fix other parameters, including the neighborhood size n and the number of Monte Carlo samples M_c, to reasonable defaults and later assess their effects via sensitivity analysis. The neighborhood size is set based on input dimension: $n = 25$ for datasets with fewer than 30 dimensions and $n = 35$ otherwise. Although (Park 2022) suggests $n \approx 15$, we find that slightly larger neighborhoods improve stability and predictive performance in higher-dimensional settings. The number of Monte Carlo samples is fixed at $M_c = 5$.

For variational optimization, we use a fixed learning rate $\eta = 0.01$ and run 300 iterations. All parameters are randomly initialized, except for the covariance matrix $\mathbf{\Sigma}_r^{(j)}$ of $\boldsymbol{r}^{(j)}$, which is initialized as $U^\top U$, where U is a randomly generated upper triangular matrix, ensuring positive definiteness and numerical stability.

5.1 Comparison to the Baseline Methods

Table 1 summarizes the average RMSE and CRPS performance of different surrogate models across various experimental settings. We also report rank scores based on both RMSE and CRPS. Note that the rank scores are computed across all repeated experiments by aggregating results from all dataset settings, thus providing a comprehensive overall ranking.

We observe from Table 1 that across all experimental configurations, DJGP consistently achieves the best rank scores in both RMSE and CRPS, highlighting its superior predictive performance and well-calibrated uncertainty estimates.

Table 1: Performance comparison of models on RMSE and CRPS.

DataSet	D	K	N	n	Feat.	Mean RMSE DGP	JGP	JGP-SIR	JGP-AE	DJGP	Mean CRPS DGP	JGP	JGP-SIR	JGP-AE	DJGP
L2	20	2	1k	25	**AE**	2.24	2.33	2.32	2.23	**2.21**	1.36	1.34	1.30	1.29	**1.29**
	20	2	1k	25	**PE**	2.33	2.23	2.32	**2.22**	2.26	1.42	**1.25**	1.31	1.26	1.27
	20	2	1k	25	**RP**	2.15	2.15	3.02	2.18	**2.15**	1.31	1.24	1.68	1.23	**1.22**
	20	2	1k	25	**RF**	**2.22**	2.26	2.41	2.24	2.27	1.34	1.28	1.34	1.26	**1.26**
LH	20	4	1k	25	**AE**	303.60	361.73	302.84	277.33	**271.39**	286.91	192.15	138.87	119.98	**114.09**
	20	4	1k	25	**PE**	314.39	290.50	**259.27**	266.15	292.97	297.39	126.00	**108.34**	111.56	117.51
	20	4	1k	25	**RP**	309.32	297.37	367.88	289.94	**283.60**	292.50	135.86	204.55	131.77	**125.55**
	20	4	1k	25	**RF**	303.19	303.00	295.86	324.80	**268.08**	287.15	137.99	127.83	162.07	**121.25**
LH	30	5	1k	25	**AE**	712.52	785.77	618.54	646.41	**563.89**	700.56	439.09	286.21	313.75	**248.25**
	30	5	1k	25	**PE**	727.80	706.05	**591.99**	636.41	604.33	714.77	347.64	**266.34**	304.36	270.85
	30	5	1k	25	**RP**	719.40	**642.72**	697.93	643.78	652.36	708.60	321.52	369.49	319.57	**317.46**
	30	5	1k	25	**RF**	711.81	711.40	611.26	656.97	**581.82**	701.22	364.36	284.54	318.57	**262.66**
LH	50	7	2k	35	**AE**	3099.28	2379.01	2488.79	2410.18	**1896.49**	3082.19	1060.76	1078.09	1044.05	**805.15**
	50	7	2k	35	**PE**	3123.36	2911.09	2497.91	2397.27	**2307.03**	3106.67	1471.66	1098.77	1022.11	**1019.60**
	50	7	2k	35	**RP**	3116.37	2508.01	2503.44	**2413.85**	2493.00	3100.04	1132.13	1115.80	**1047.43**	1099.38
	50	7	2k	35	**RF**	3109.00	2729.22	2482.38	2412.44	**2379.80**	3092.08	1300.04	1099.84	1047.86	**1011.23**
RankScore						4.13	3.38	3.19	2.44	**1.88**	4.88	3.44	3.06	2.13	**1.50**

Note. Smaller values indicate better predictive performance. Here, D denotes the input dimension, K the latent dimensionality, N the total number of training samples, and n the local neighborhood size used in JGP-based methods.

Notably, DJGP consistently outperforms JGP, its dimension-reduced variants (JGP-SIR and JGP-AE), and DGP in both RMSE and CRPS. While JGP-SIR improves upon JGP, the gains are limited under random projection (RP), where JGP can already adapt well to linear structure via lengthscale learning. In contrast, JGP-AE often performs better by capturing nonlinear mappings and preserving richer latent information. Overall, DJGP achieves the best performance in most settings and remains competitive elsewhere, demonstrating robust effectiveness across diverse scenarios.

DGPs perform comparatively poorly, particularly in terms of CRPS, indicating overconfident predictions and weak uncertainty calibration. This reflects a key distinction between global and local models: DGPs learn a single mapping over the entire input space, which limits adaptability in regions with abrupt changes, whereas local models such as JGP and DJGP can better capture such heterogeneity. Moreover, DGPs typically require large datasets to effectively learn hierarchical representations; with only $N = 1000$ training points, their performance is constrained.

In summary, DJGP offers a compelling balance between accuracy and uncertainty estimation, and its robustness across various datasets and feature transformations demonstrates its effectiveness for high-dimensional, piecewise continuous surrogate modeling.

5.2 Performance Under Different Latent Dimension K, Observed Dimension D, and Data Size N

To gain a deeper understanding of DJGP's behavior under varying data conditions, we evaluate its predictive performance across different configurations. Specifically, we vary the latent dimension K, the observed (expanded) dimension D, and the number of training samples N, while fixing $J = 100$ and $n = 35$. All experiments are conducted using the LH dataset with the RF expansion; similar trends are observed with other datasets.

Figure 1 presents RMSE results for varying observed dimensions D, with fixed latent dimension $K = 5$ and training sizes $N \in \{1000, 3000, 5000\}$. RMSE remains largely stable as D increases, indicating that DJGP performs effective dimension reduction without substantial loss of predictive accuracy even in high-dimensional observed spaces. This weak dependence on D is consistent with Theorem 1. Although the projection approximation term contains the factor $KD\,L_2^{-1}$, the dominant statistical term in the risk is $\mathrm{GP}_{\mathrm{oracle}}(n, K)$, which depends only on the intrinsic latent dimension K rather than the ambient input dimension D. The empirical stability of RMSE therefore suggests that the projection-related terms remain controlled in this setting, so that prediction error is primarily governed by latent-space GP estimation rather than by ambient dimensionality.

Figure 2 shows RMSE as N increases with fixed $K \in \{3, 5\}$ and $D = 30$. Although the neighborhood size n is fixed, increasing the global sample size makes the local neighborhood around each test point denser, thereby reducing the neighborhood radius $\rho_r(\boldsymbol{x}_*)$. According to Theorem 1, this directly reduces the local linearization term $\mathbb{E}[\rho_r(\boldsymbol{X})^4]$, which arises from the Taylor remainder of approximating $g(\cdot)$ within each neighborhood. A smaller neighborhood also reduces local heterogeneity near region boundaries, which helps control the gating contribution $(\tau^2 + \tau^{-1}\epsilon_n)\Delta_f^2$. These effects together explain the monotone improvement in RMSE as N increases, even when the local sample size n remains fixed.

We also examined the RMSE trend of DJGP with varying latent dimensionality $K \in \{3, \ldots, 8\}$ and training sizes $N \in \{1000, 3000, 5000, 10000\}$, while keeping the original input dimension fixed at $D = 30$. Figure 3 (left) presents the mean RMSE as a function of N, K, while the right panel shows an approximately linear relationship between log(RMSE) and $\log(N^K)$. Equivalently, RMSE exhibits an approximate power-law dependence on N whose exponent scales with the latent dimension K, indicating that the effective learning complexity is governed by K rather than the ambient dimension D. This trend is consistent with theoretical bounds for GP regression where the rate depends on the effective input dimension, suggesting that DJGP effectively estimates the latent dimensionality and model nonstationarity and discontinuity in the data effectively.

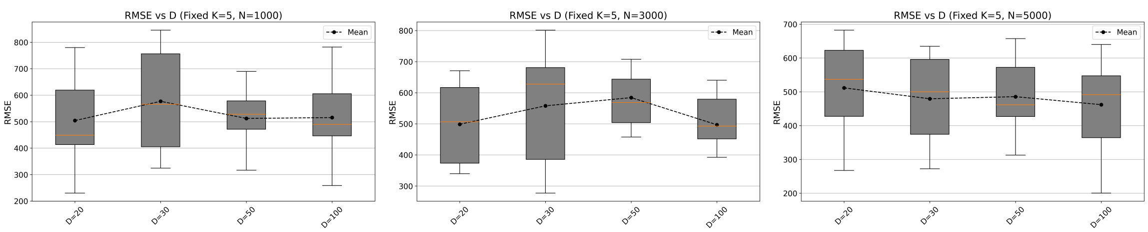

Figure 1: Effect of observed dimension D on RMSE under fixed latent dimension $K = 5$ and varying training sizes. DJGP maintains stable performance even with increased dimensionality.

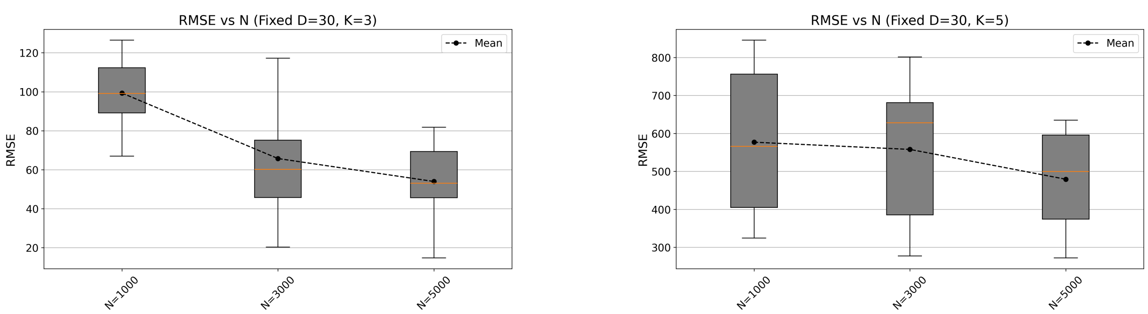

Figure 2: Effect of training set size N on RMSE under fixed observed dimension $D = 30$ and different latent dimensions. Larger N leads to denser local neighborhoods and improved performance.

5.3 Sensitivity to the Tuning Parameters

Selecting tuning parameters such as the numbers of inducing points (L_1, L_2), the neighborhood size n, and the target latent dimension Q is an important practical issue for DJGP.

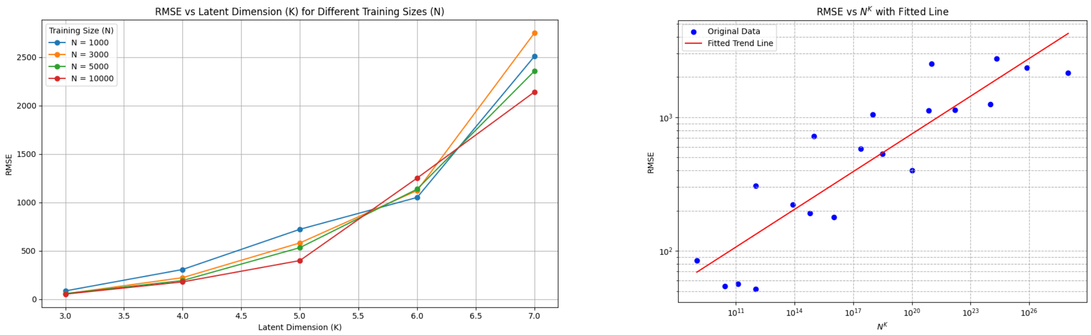

Figure 3: Effect of latent dimension K on DJGP performance. (Left) Mean RMSE as a function of the latent dimension K under varying sample sizes N. (Right) RMSE exhibits a scaling trend governed by N^K rather than N^D, consistent with the theoretical error behavior of stationary Gaussian processes with intrinsic input dimension K(Park 2022). This indicates that DJGP effectively identifies the latent subspace and mitigates the curse of dimensionality in high-dimensional observations.

In this section, we summarize a sensitivity analysis on the LH dataset with RF expansion.

5.3.1 Influence of Neighborhood Size and Inducing Points

We first examine the sensitivity of DJGP to the numbers of local and global inducing points (L_1, L_2) and to the neighborhood size n. The experiments are conducted on the LH dataset with fixed $(D, K, N, J) = (30, 5, 1000, 100)$. We vary (L_1, L_2) over the grid $\{2, 4, 6\} \times \{20, 40, 60\}$ and evaluate the resulting RMSE. The corresponding CRPS values exhibit the same overall trend and therefore lead to the same practical conclusions.

Figure 4(a) shows the joint effect of (L_1, L_2) on predictive performance. We observe that $(L_1, L_2) = (4, 40)$ provides the best overall trade-off between predictive accuracy and computational cost. When L_1 is too small, the local variational approximation starts to be too rough, whereas large L_1 can lead to over-parameterization relative to the limited local sample size n, which degrades generalization errors. Increasing L_2 tends to improve performance more consistently, as a larger global inducing set provides a better approximation to the nonlinear projection structure.

Figure 4(b) further shows the joint influence of neighborhood size n and local inducing count L_1. Moderate neighborhood sizes perform best: larger neighborhoods provide more data for estimation but increase Taylor-approximation errors, especially in datasets with sharp discontinuities. In our experiments, neighborhood sizes between 15 and 35 generally yield favorable results, depending on the local complexity of the data.

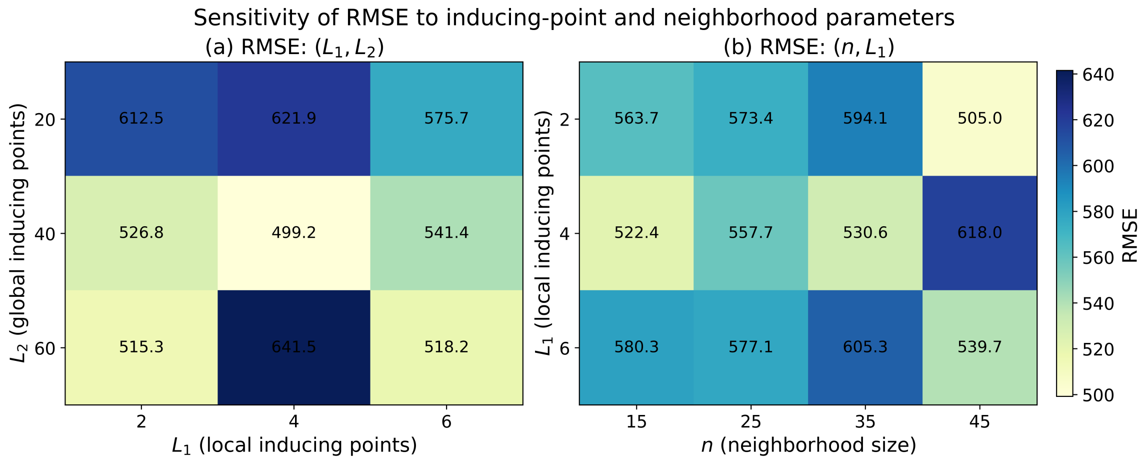

Figure 4: Sensitivity of RMSE to key hyperparameters. Left: joint effect of the local and global inducing point counts (L_1, L_2). Right: joint effect of neighborhood size n and local inducing point count L_1.

5.3.2 Influence of the Selected Latent Dimension Q

We again use the LH dataset and fix $N = 1000$, $D = 30$, and $n = 35$. We evaluate DJGP under combinations of $K \in \{2, 3, 5, 7\}$ and $Q \in \{2, 3, 5, 7\}$ and report the resulting RMSE in Figure 5.

Across settings, $Q = 3$ or $Q = 5$ often provides strong and stable performance. Too small Q causes over-simplified latent structure, degrading prediction accuracy, while too large Q complicates local partitioning and downstream JGP modeling, promoting overfitting. Interestingly, the best-performing value of Q does not always coincide with the ground-truth intrinsic dimension K. This suggests that the target latent dimension is best treated as a tuning parameter rather than fixed to the presumed intrinsic dimension.

5.3.3 Practical Guidance

Based on the sensitivity analysis and the benchmark experiments, we recommend using

$$n \in [25, 35], \qquad (L_1, L_2) = (4, 40), \qquad Q = 5$$

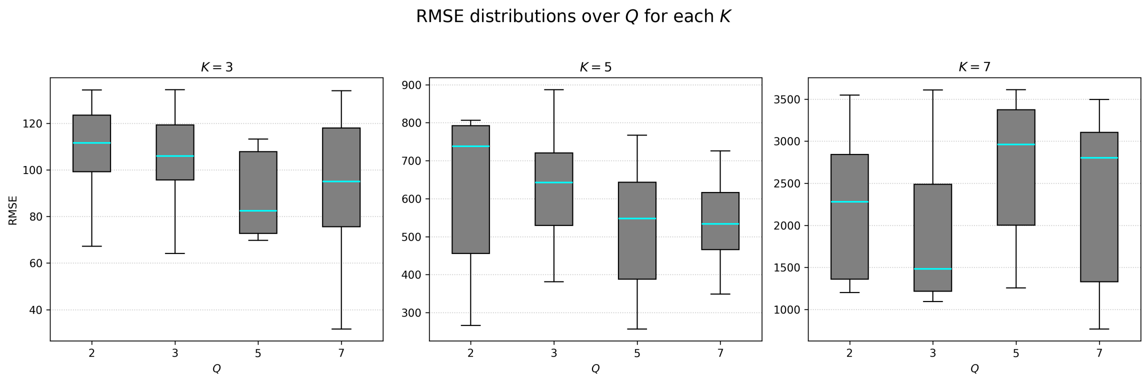

Figure 5: Effect of the target latent dimension Q on RMSE across different ground-truth latent dimensions K, evaluated on the LH dataset using RF expansion. Each box summarizes RMSE over 10 randomized train–test splits.

as a practical default configuration for DJGP. These values provide a good balance between predictive performance and computational cost across the datasets considered in this paper.

When additional computational budget is available, these defaults can be further refined by cross-validation or a held-out validation set. In particular, we find it useful to explore $Q \in [3, 7]$, as performance gains typically plateau once Q becomes larger.

6 Real Dataset Experiments

We evaluate DJGP and baseline models on three benchmark datasets from the UCI repository: Wine Quality, Parkinson's Telemonitoring, and Appliances Energy Prediction. These datasets were chosen to span a range of sample sizes, input dimensions, and, importantly, response surface roughness (i.e. degree of response jumps). Table 2 summarizes their main characteristics, including training size N, input dimension D, test size J, and the latent dimension K used in the model.

To characterize local response complexity, we report three summary measures: the average local gradient G_a, the maximum local gradient G_m, and a projection-based roughness measure TV_2. These quantities summarize average local variation, worst-case local irregularity, and overall response roughness of the regression surface, respectively.

We use test sets of approximately 10% of the data for the smaller datasets (Wine Quality and Parkinson's). For the larger Appliances dataset (approximately 20,000 samples), we fix the test set size at about 600 points to maintain comparable computational cost across methods. Since DJGP and its baselines rely on local or Monte Carlo-based inference at each test input, prediction cost increases roughly linearly with the number of test points. All dataset splits are repeated over 10 random seeds, and results are averaged across runs.

Unless otherwise specified, the latent dimension K for each method is selected by five-fold cross-validation on one random split and then fixed for the remaining splits. For DGP, K denotes the latent dimension of the final hidden layer. All models are trained with learning rate 0.01, with early stopping based on a 10% validation subset of the training data. For DJGP, we use $(L_1, L_2) = (4, 40)$, neighborhood size $n = 35$, and Monte Carlo sample size $M_c = 3$, following Section 5.3.

Table 2: Dataset statistics and smoothness metrics.

Dataset	N	D	J	K	G_a	G_m	TV_2
Wine Quality	6,497	11	650	3	0.327	9.90	8,798
Parkinson's Telemonitoring	5,875	19	588	5	2.685	56.83	60,872
Appliances Energy Prediction	19,735	28	593	5	6.459	355.17	2,849,220

1. $G_a = \frac{1}{|\mathcal{E}|}\sum_{(i,j)\in\mathcal{E}} \frac{|y_i - y_j|}{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|}$: average local gradient magnitude over the edge set $\mathcal{E}$ of a k-nearest neighbor graph ($k = 6$) constructed on the training inputs.

2. $G_m = \max_{(i,j)\in\mathcal{E}} \frac{|y_i - y_j|}{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|}$: maximum local gradient magnitude over the same graph, summarizing worst-case local irregularity.

3. $\mathrm{TV}_2 = \sum_{i=2}^{N-1} \left|(y_{(i+1)} - y_{(i)}) - (y_{(i)} - y_{(i-1)})\right|$: projection-based roughness measure computed after projecting the inputs onto the first principal component and ordering the responses accordingly.

Figures 6 show the distribution of RMSE and CRPS across 10 randomized train–test splits. Overall, JGP-based models consistently outperform DGP, supporting the hypothesis that local models are better suited for handling heterogeneous structures and potential discontinuities. DJGP consistently achieves the best performance across all datasets, excelling in both RMSE and CRPS. On the **Wine Quality** dataset, DJGP achieves the lowest median RMSE while the RMSE metric has relatively higher variations–attributable to the stochasticity introduced by sampling latent projection matrices during inference. On the **Parkinson's Telemonitoring** dataset, where PCA, SIR, and AE all degrade the performance of vanilla JGP, DJGP significantly outperforms all baselines. This highlights the advantage of its integrated dimensionality reduction, which avoids the information loss often introduced

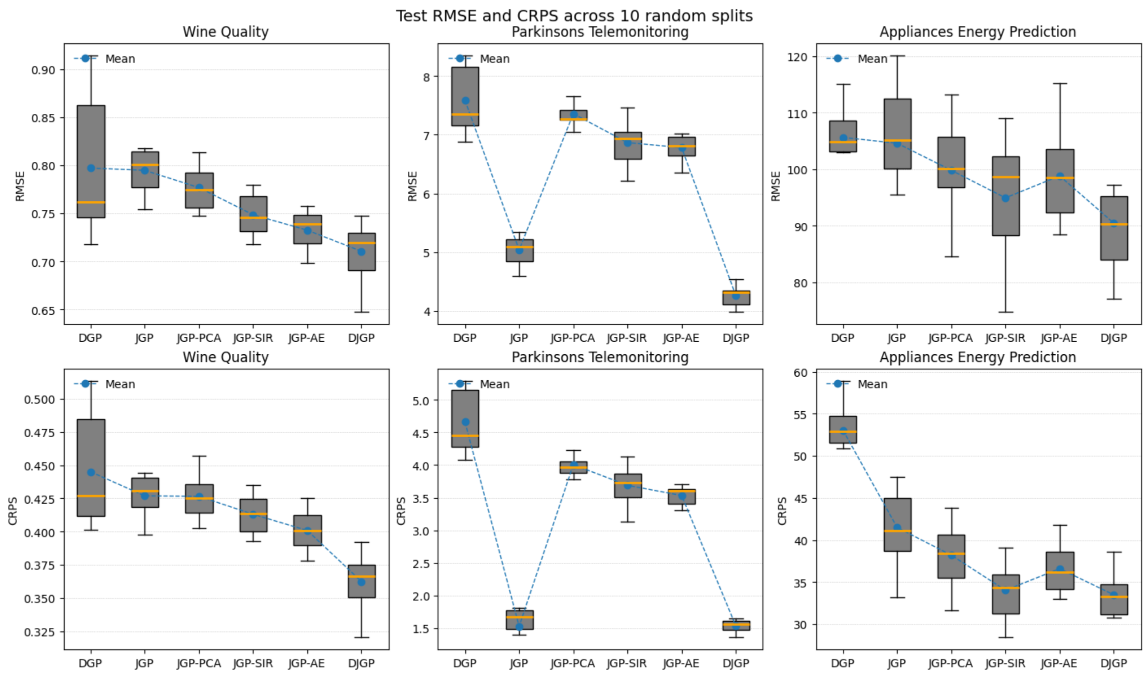

Figure 6: Predictive performance comparison across real datasets.

by two-stage projection methods. On the most challenging dataset—**Appliances Energy Prediction**—which is both high-dimensional and structurally irregular, DJGP delivers substantial performance gains, demonstrating its scalability and robustness in large-scale, complex regression settings.

DGP, although theoretically capable of modeling nonstationarity, exhibits the weakest performance overall. Its underperformance is likely due to a mismatch between its smooth functional assumptions and the presence of jump discontinuities or outliers. DGP also shows higher variability on the smaller Wine and Parkinson's datasets, suggesting it is more data-hungry and less robust in low-data regimes.

Comparison of the computation times among the compared methods are placed in Appendix D.

7 Conclusion

We introduced the Deep Jump Gaussian Process (DJGP), a surrogate model that unifies global subspace learning with local discontinuity detection. By placing Gaussian process priors on region-specific projection matrices and integrating dimension reduction into JGP, DJGP jointly learns low-dimensional representations and piecewise-continuous regimes in high-dimensional inputs. A gradient-based variational inference algorithm enables joint optimization of projection parameters, local JGP hyperparameters, and partitioning structure, while inducing-point approximations ensure scalability.

On the theoretical side, we established an oracle bound that decomposes the DJGP prediction error into contributions from mis-gating, projection estimation, local linearization, and latent-space GP estimation, providing insight into the sources of predictive accuracy.

Empirically, across synthetic benchmarks and three real-world UCI datasets, DJGP consistently achieves lower RMSE and CRPS than competing methods, including JGP (with and without dimension reduction) and deep GPs. Its integrated dimensionality reduction mitigates overfitting in local neighborhoods and improves predictive reliability in sparse, high-dimensional settings.

DJGP's ability to capture abrupt regime changes while providing uncertainty quantification makes it well suited for applications such as materials science, econometrics, and social sciences, where structural shifts are common.

Data Availability Statement

To support reproducibility, the synthetic and real datasets used in this study are shared along with the source code.

Conflict of Interest Statement

The authors report there are no competing interests to declare.

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